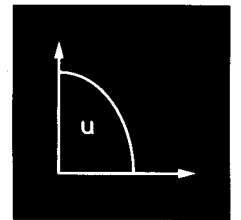
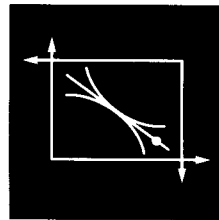
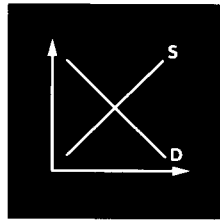
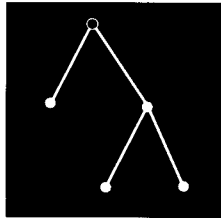
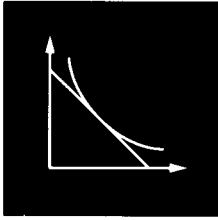


Microeconomic Theory



Microeconomic Theory

Andreu Mas-Colell Michael D. Whinston

and

Jerry R. Green

OXFORD UNIVERSITY PRESS

Oxford New York
Athens Auckland Bangkok Bombay
Calcutta Cape Town Dar es Salaam Delhi
Florence Hong Kong Istanbul Karachi
Kuala Lumpur Madras Madrid Melbourne
Mexico City Nairobi Paris Singapore
Taipei Tokyo Toronto

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Published by Oxford University Press, Inc.,
198 Madison Avenue, New York, New York 10016

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Library of Congress Cataloging-in-Publication Data

Mas-Colell, Andreu.

Microeconomic theory / Andreu Mas-Colell, Michael D. Whinston, and Jerry R. Green.
p. cm.

Includes bibliographical references and index.

ISBN 0-19-507340-1 (cloth)

ISBN 0-19-510268-1 (paper)

I. Microeconomics. I. Whinston, Michael Dennis. II. Green, Jerry R. III. Title.

HB172.M6247 1995

338.5—dc20 95-18128

9 8 7 6 5

Printed in the United States of America
on acid-free paper

A Esther, por todo A. M.-C.

To Bonnie, for keeping me smiling throughout;
to Noah, for his sweetness and joy at the book's completion;
and to Nan, for helping me get started M. D. W.

To Pamela, for her kindness, character, strength, and spirit J. R. G.

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Preface

Microeconomic Theory is intended to serve as the text for a first-year graduate course in microeconomic theory. The original sources for much of the book's material are the lecture notes that we have provided over the years to students in the first-year microeconomic theory course at Harvard. Starting from these notes, we have tried to produce a text that covers in an accessible yet rigorous way the full range of topics taught in a typical first-year course.

The nonlexicographic ordering of our names deserves some explanation. The project was first planned and begun by the three of us in the spring of 1990. However, in February 1992, after early versions of most of the book's chapters had been drafted, Jerry Green was selected to serve as Provost of Harvard University, a position that forced him to suspend his involvement in the project. From this point in time until the manuscript's completion in June 1994, Andreu Mas-Colell and Michael Whinston assumed full responsibility for the project. With the conclusion of Jerry Green's service as Provost, the original three-person team was reunited for the review of galley and page proofs during the winter of 1994/1995.

The Organization of the Book

Microeconomic theory as a discipline begins by considering the behavior of individual agents and builds from this foundation to a theory of aggregate economic outcomes. *Microeconomic Theory* (the book) follows exactly this outline. It is divided into five parts. Part I covers individual decision making. It opens with a general treatment of individual choice and proceeds to develop the classical theories of consumer and producer behavior. It also provides an introduction to the theory of individual choice under uncertainty. Part II covers game theory, the extension of the theory of individual decision making to situations in which several decision makers interact. Part III initiates the investigation of market equilibria. It begins with an introduction to competitive equilibrium and the fundamental theorems of welfare economics in the context of the Marshallian partial equilibrium model. It then explores the possibilities for market failures in the presence of externalities, market power, and asymmetric information. Part IV substantially extends our previous study of competitive markets to the general equilibrium context. The positive and normative aspects of the theory are examined in detail, as are extensions of the theory to equilibrium under uncertainty and over time. Part V studies welfare economics. It discusses the possibilities for aggregation of individual preferences into social preferences both with and without interpersonal utility comparisons, as well as the implementation of social choices in the presence of incomplete information about agents' preferences. A Mathematical Appendix provides an introduction to most of the more advanced mathematics used in the book (e.g., concave/convex

functions, constrained optimization techniques, fixed point theorems, etc.) as well as references for further reading.

The Style of the Book

In choosing the content of *Microeconomic Theory* we have tried to err on the side of inclusion. Our aim has been to assure coverage of most topics that instructors in a first-year graduate microeconomic theory course might want to teach. An inevitable consequence of this choice is that the book covers more topics than any single first-year course can discuss adequately. (We certainly have never taught all of it in any one year.) Our hope is that the range of topics presented will allow instructors the freedom to emphasize those they find most important.

We have sought a style of presentation that is accessible, yet also rigorous. Wherever possible we give precise definitions and formal proofs of propositions. At the same time, we accompany this analysis with extensive verbal discussion as well as with numerous examples to illustrate key concepts. Where we have considered a proof or topic either too difficult or too peripheral we have put it into smaller type to allow students to skip over it easily in a first reading.

Each chapter offers many exercises, ranging from easy to hard [graded from A (easiest) to C (hardest)] to help students master the material. Some of these exercises also appear within the text of the chapters so that students can check their understanding along the way (almost all of these are level A exercises).

The mathematical prerequisites for use of the book are a basic knowledge of calculus, some familiarity with linear algebra (although the use of vectors and matrices is introduced gradually in Part I), and a grasp of the elementary aspects of probability. Students also will find helpful some familiarity with microeconomics at the level of an intermediate undergraduate course.

Teaching the Book

The material in this book may be taught in many different sequences. Typically we have taught Parts I–III in the Fall semester and Parts IV and V in the Spring (omitting some topics in each case). A very natural alternative to this sequence (one used in a number of departments that we know of) might instead teach Parts I and IV in the Fall, and Parts II, III, and V in the Spring.¹ The advantage of this alternative sequence is that the study of general equilibrium analysis more closely follows the study of individual behavior in competitive markets that is developed in Part I. The disadvantage, and the reason we have not used this sequence in our own course, is that this makes for a more abstract first semester; our students have seemed happy to have the change of pace offered by game theory, oligopoly, and asymmetric information after studying Part I.

The chapters have been written to be relatively self-contained. As a result, they can be shifted easily among the parts to accommodate many other course sequences. For example, we have often opted to teach game theory on an “as needed” basis,

1. Obviously, some adjustment needs to be made by programs that operate on a quarter system.

breaking it up into segments that are discussed right before they are used (e.g., Chapter 7, Chapter 8, and Sections 9.A–B before studying oligopoly, Sections 9.C–D before covering signaling). Some other possibilities include teaching the aggregation of preferences (Chapter 21) immediately after individual decision making and covering the principal-agent problem (Chapter 14), adverse selection, signaling, and screening (Chapter 13), and mechanism design (Chapter 23) together in a section of the course focusing on information economics.

In addition, even within each part, the sequence of topics can often be altered easily. For example, it has been common in many programs to teach the preference-based theory of consumer demand before teaching the revealed preference, or “choice-based,” theory. Although we think there are good reasons to reverse this sequence as we have done in Part I,² we have made sure that the material on demand can be covered in this more traditional way as well.³

On Mathematical Notation

For the most part, our use of mathematical notation is standard. Perhaps the most important mathematical rule to keep straight regards matrix notation. Put simply, vectors are always treated mathematically as *column vectors*, even though they are often displayed within the written text as rows to conserve space. The transpose of the (column) vector x is denoted by x^T . When taking the inner product of two (column) vectors x and y , we write $x \cdot y$; it has the same meaning as $x^T y$. This and other aspects of matrix notation are reviewed in greater detail in Section M.A of the Mathematical Appendix.

To help highlight definitions and propositions we have chosen to display them in a different typeface than is used elsewhere in the text. One perhaps unfortunate consequence of this choice is that mathematical symbols sometimes appear slightly differently there than in the rest of the text. With this warning, we hope that no confusion will result.

Summation symbols ($\sum$) are displayed in various ways throughout the text. Sometimes they are written as

$$\sum_{n=1}^N$$

(usually only in displayed equations), but often to conserve space they appear as $\sum_{n=1}^N$, and in the many cases in which no confusion exists about the upper and lower limit of the index in the summation, we typically write just $\sum_n$. A similar point applies to the product symbol $\prod$.

2. In particular, it is *much* easier to introduce and derive many properties of demand in the choice-based theory than it is using the preference-based approach; and the choice-based theory gives you *almost* all the properties of demand that follow from assuming the existence of rational preferences.

3. To do this, one introduces the basics of the consumer's problem using Sections 2.A–D and 3.A–D, discusses the properties of uncompensated and compensated demand functions, the indirect utility function, and the expenditure function using Sections 3.D–I and 2.E, and then studies revealed preference theory using Sections 2.F and 3.J (and Chapter 1 for a more general overview of the two approaches).

Also described below are the meanings we attach to a few mathematical symbols whose use is somewhat less uniform in the literature [in this list, $x = (x_1, \dots, x_N)$ and $y = (y_1, \dots, y_N)$ are (column) vectors, while X and Y are sets]:

<i>Symbol</i>	<i>Meaning</i>
$x \geq y$	$x_n \geq y_n$ for all $n = 1, \dots, N$.
$x \gg y$	$x_n > y_n$ for all $n = 1, \dots, N$.
$X \subset Y$	weak set inclusion ($x \in X$ implies $x \in Y$).
$X \setminus Y$	The set $\{x: x \in X \text{ but } x \notin Y\}$.
$E_x[f(x, y)]$	The expected value of the function $f(\cdot)$ over realizations of the random variable x . (When the expectation is over all of the arguments of the function we simply write $E[f(x, y)]$.)

Acknowledgments

Many people have contributed to the development of this book. Dilip Abreu, Doug Bernheim, David Card, Prajit Dutta, Steve Goldman, John Panzar, and David Pearce all (bravely) test-taught a very early version of the manuscript during the 1991–92 academic year. Their comments at that early stage were instrumental in the refinement of the book into its current style, and led to many other substantive improvements in the text. Our colleagues (and in some cases former students) Luis Corchón, Simon Grant, Drew Fudenberg, Chiaki Hara, Sergiu Hart, Bengt Holmstrom, Eric Maskin, John Nachbar, Martin Osborne, Ben Polak, Ariel Rubinstein, and Martin Weitzman offered numerous helpful suggestions. The book would undoubtedly have been better still had we managed to incorporate all of their ideas.

Many generations of first-year Harvard graduate students have helped us with their questions, comments, and corrections. In addition, a number of current and former students have played a more formal role in the book's development, serving as research assistants in various capacities. Shira Lewin read the entire manuscript, finding errors in our proofs, suggesting improvements in exposition, and even (indeed, often) correcting our grammar. Chiaki Hara, Ilya Segal, and Steve Tadelis, with the assistance of Marc Nachman, have checked that the book's many exercises could be solved, and have suggested how they might be formulated properly when our first attempt to do so failed. Chiaki Hara and Steve Tadelis have also given us extensive comments and corrections on the text itself. Emily Mechner, Nick Palmer, Phil Panet, and Billy Pizer were members of a team of first-year students that read our early drafts in the summer of 1992 and offered very helpful suggestions on how we could convey the material better.

Betsy Carpenter and Claudia Napolilli provided expert secretarial support throughout the project, helping to type some chapter drafts, copying material on very tight deadlines, and providing their support in hundreds of other ways. Gloria Gerrig kept careful track of our ever-increasing expenditures.

Our editor at Oxford, Herb Addison, was instrumental in developing the test teaching program that so helped us in the book's early stages, and offered his support throughout the book's development. Leslie Phillips of Oxford took our expression of appreciation for the look of the Feynman Lectures, and turned it into a book design that exceeded our highest expectations. Alan Chesterton and the rest of the

staff at Keyword Publishing Services did an absolutely superb job editing and producing the book on a very tight schedule. Their complete professionalism has been deeply appreciated.

The influence of many other individuals on the book, although more indirect, has been no less important. Many of the exercises that appear in the book have been conceived over the years by others, both at Harvard and elsewhere. We have indicated our source for an exercise whenever we were aware of it. Good exercises are an enormously valuable resource. We thank the anonymous authors of many of the exercises that appear here.

The work of numerous scholars has contributed to our knowledge of the topics discussed in this book. Of necessity we have been able to provide references in each chapter to only a limited number of sources. Many interesting and important contributions have not been included. These usually can be found in the references of the works we do list; indeed, most chapters include at least one reference to a general survey of their topic.

We have also had the good fortune to teach the first-year graduate microeconomic theory course at Harvard in the years prior to writing this book with Ken Arrow, Dale Jorgenson, Steve Marglin, Eric Maskin, and Mike Spence, from whom we learned a great deal about microeconomics and its teaching.

We also thank the NSF and Sloan Foundation for their support of our research over the years. In addition, the Center for Advanced Study in the Behavioral Sciences provided an ideal environment to Michael Whinston for completing the manuscript during the 1993/1994 academic year. The Universitat Pompeu Fabra also offered its hospitality to Andreu Mas-Colell at numerous points during the book's development.

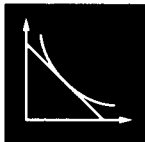
Finally, we want to offer a special thanks to those who first excited us about the subject matter that appears here: Gerard Debreu, Leo Hurwicz, Roy Radner, Marcel Richter, and Hugo Sonnenschein (A.M.-C.); David Cass, Peter Diamond, Franklin Fisher, Sanford Grossman, and Eric Maskin (M.D.W.); Emmanuel Drandakis, Ron Jones, Lionel McKenzie, and Edward Zabel (J.R.G.).

A.M.-C., M.D.W., J.R.G.

Cambridge, MA

March 1995

Microeconomic Theory



Individual Decision Making

A distinctive feature of microeconomic theory is that it aims to model economic activity as an interaction of individual economic agents pursuing their private interests. It is therefore appropriate that we begin our study of microeconomic theory with an analysis of individual decision making.

Chapter 1 is short and preliminary. It consists of an introduction to the theory of individual decision making considered in an abstract setting. It introduces the decision maker and her choice problem, and it describes two related approaches to modeling her decisions. One, the *preference-based approach*, assumes that the decision maker has a preference relation over her set of possible choices that satisfies certain rationality axioms. The other, the *choice-based approach*, focuses directly on the decision maker's choice behavior, imposing consistency restrictions that parallel the rationality axioms of the preference-based approach.

The remaining chapters in Part One study individual decision making in explicitly economic contexts. It is common in microeconomics texts—and this text is no exception—to distinguish between two sets of agents in the economy: *individual consumers* and *firms*. Because individual consumers own and run firms and therefore ultimately determine a firm's actions, they are in a sense the more fundamental element of an economic model. Hence, we begin our review of the theory of economic decision making with an examination of the consumption side of the economy.

Chapters 2 and 3 study the behavior of consumers in a market economy. Chapter 2 begins by describing the consumer's decision problem and then introduces the concept of the consumer's *demand function*. We then proceed to investigate the implications for the demand function of several natural properties of consumer demand. This investigation constitutes an analysis of consumer behavior in the spirit of the choice-based approach introduced in Chapter 1.

In Chapter 3, we develop the classical preference-based approach to consumer demand. Topics such as utility maximization, expenditure minimization, duality, integrability, and the measurement of welfare changes are studied there. We also discuss the relation between this theory and the choice-based approach studied in Chapter 2.

In economic analysis, the aggregate behavior of consumers is often more important than the behavior of any single consumer. In Chapter 4, we analyze the

extent to which the properties of individual demand discussed in Chapters 2 and 3 also hold for aggregate consumer demand.

In Chapter 5, we study the behavior of the firm. We begin by posing the firm's decision problem, introducing its technological constraints and the assumption of profit maximization. A rich theory, paralleling that for consumer demand, emerges. In an important sense, however, this analysis constitutes a first step because it takes the objective of profit maximization as a maintained hypothesis. In the last section of the chapter, we comment on the circumstances under which profit maximization can be derived as the desired objective of the firm's owners.

Chapter 6 introduces risk and uncertainty into the theory of individual decision making. In most economic decision problems, an individual's or firm's choices do not result in perfectly certain outcomes. The theory of decision making under uncertainty developed in this chapter therefore has wide-ranging applications to economic problems, many of which we discuss later in the book.

Preference and Choice

1.A Introduction

In this chapter, we begin our study of the theory of individual decision making by considering it in a completely abstract setting. The remaining chapters in Part I develop the analysis in the context of explicitly economic decisions.

The starting point for any individual decision problem is a *set of possible (mutually exclusive) alternatives* from which the individual must choose. In the discussion that follows, we denote this set of alternatives abstractly by X . For the moment, this set can be anything. For example, when an individual confronts a decision of what career path to follow, the alternatives in X might be: {go to law school, go to graduate school and study economics, go to business school, . . . , become a rock star}. In Chapters 2 and 3, when we consider the consumer's decision problem, the elements of the set X are the possible consumption choices.

There are two distinct approaches to modeling individual choice behavior. The first, which we introduce in Section 1.B, treats the decision maker's tastes, as summarized in her *preference relation*, as the primitive characteristic of the individual. The theory is developed by first imposing rationality axioms on the decision maker's preferences and then analyzing the consequences of these preferences for her choice behavior (i.e., on decisions made). This preference-based approach is the more traditional of the two, and it is the one that we emphasize throughout the book.

The second approach, which we develop in Section 1.C, treats the individual's choice behavior as the primitive feature and proceeds by making assumptions directly concerning this behavior. A central assumption in this approach, the *weak axiom of revealed preference*, imposes an element of consistency on choice behavior, in a sense paralleling the rationality assumptions of the preference-based approach. This choice-based approach has several attractive features. It leaves room, in principle, for more general forms of individual behavior than is possible with the preference-based approach. It also makes assumptions about objects that are directly observable (choice behavior), rather than about things that are not (preferences). Perhaps most importantly, it makes clear that the theory of individual decision making need not be based on a process of introspection but can be given an entirely behavioral foundation.

Understanding the relationship between these two different approaches to modeling individual behavior is of considerable interest. Section 1.D investigates this question, examining first the implications of the preference-based approach for choice behavior and then the conditions under which choice behavior is compatible with the existence of underlying preferences. (This is an issue that also comes up in Chapters 2 and 3 for the more restricted setting of consumer demand.)

For an in-depth, advanced treatment of the material of this chapter, see Richter (1971).

1.B Preference Relations

In the preference-based approach, the objectives of the decision maker are summarized in a *preference relation*, which we denote by $\succsim$. Technically, $\succsim$ is a binary relation on the set of alternatives X , allowing the comparison of pairs of alternatives $x, y \in X$. We read $x \succsim y$ as “ x is at least as good as y .” From $\succsim$, we can derive two other important relations on X :

- (i) The *strict preference* relation, $\succ$, defined by

$$x \succ y \Leftrightarrow x \succsim y \text{ but not } y \succsim x$$

and read “ x is preferred to y .”¹

- (ii) The *indifference* relation, $\sim$, defined by

$$x \sim y \Leftrightarrow x \succsim y \text{ and } y \succsim x$$

and read “ x is indifferent to y .”

In much of microeconomic theory, individual preferences are assumed to be *rational*. The hypothesis of rationality is embodied in two basic assumptions about the preference relation $\succsim$: *completeness* and *transitivity*.²

Definition 1.B.1: The preference relation $\succsim$ is *rational* if it possesses the following two properties:

- (i) *Completeness:* for all $x, y \in X$, we have that $x \succsim y$ or $y \succsim x$ (or both).
- (ii) *Transitivity:* For all $x, y, z \in X$, if $x \succsim y$ and $y \succsim z$, then $x \succsim z$.

The assumption that $\succsim$ is complete says that the individual has a well-defined preference between any two possible alternatives. The strength of the completeness assumption should not be underestimated. Introspection quickly reveals how hard it is to evaluate alternatives that are far from the realm of common experience. It takes work and serious reflection to find out one’s own preferences. The completeness axiom says that this task has taken place: our decision makers make only meditated choices.

Transitivity is also a strong assumption, and it goes to the heart of the concept of

1. The symbol $\Leftrightarrow$ is read as “if and only if.” The literature sometimes speaks of $x \succsim y$ as “ x is weakly preferred to y ” and $x \succ y$ as “ x is strictly preferred to y .” We shall adhere to the terminology introduced above.

2. Note that there is no unified terminology in the literature; *weak order* and *complete preorder* are common alternatives to the term *rational preference relation*. Also, in some presentations, the assumption that $\succsim$ is *reflexive* (defined as $x \succsim x$ for all $x \in X$) is added to the completeness and transitivity assumptions. This property is, in fact, implied by completeness and so is redundant.

rationality. Transitivity implies that it is impossible to face the decision maker with a sequence of pairwise choices in which her preferences appear to cycle: for example, feeling that an apple is at least as good as a banana and that a banana is at least as good as an orange but then also preferring an orange over an apple. Like the completeness property, the transitivity assumption can be hard to satisfy when evaluating alternatives far from common experience. As compared to the completeness property, however, it is also more fundamental in the sense that substantial portions of economic theory would not survive if economic agents could not be assumed to have transitive preferences.

The assumption that the preference relation $\succsim$ is complete and transitive has implications for the strict preference and indifference relations $\succ$ and $\sim$. These are summarized in Proposition 1.B.1, whose proof we forgo. (After completing this section, try to establish these properties yourself in Exercises 1.B.1 and 1.B.2.)

Proposition 1.B.1: If $\succsim$ is rational then:

- (i) $\succ$ is both *irreflexive* ($x \succ x$ never holds) and *transitive* (if $x \succ y$ and $y \succ z$, then $x \succ z$).
- (ii) $\sim$ is *reflexive* ($x \sim x$ for all x), *transitive* (if $x \sim y$ and $y \sim z$, then $x \sim z$), and *symmetric* (if $x \sim y$, then $y \sim x$).
- (iii) if $x \succ y \succsim z$, then $x \succ z$.

The irreflexivity of $\succ$ and the reflexivity and symmetry of $\sim$ are sensible properties for strict preference and indifference relations. A more important point in Proposition 1.B.1 is that rationality of $\succsim$ implies that both $\succ$ and $\sim$ are transitive. In addition, a transitive-like property also holds for $\succ$ when it is combined with an at-least-as-good-as relation, $\succsim$.

An individual's preferences may fail to satisfy the transitivity property for a number of reasons. One difficulty arises because of the problem of *just perceptible differences*. For example, if we ask an individual to choose between two very similar shades of gray for painting her room, she may be unable to tell the difference between the colors and will therefore be indifferent. Suppose now that we offer her a choice between the lighter of the two gray paints and a slightly lighter shade. She may again be unable to tell the difference. If we continue in this fashion, letting the paint colors get progressively lighter with each successive choice experiment, she may express indifference at each step. Yet, if we offer her a choice between the original (darkest) shade of gray and the final (almost white) color, she would be able to distinguish between the colors and is likely to prefer one of them. This, however, violates transitivity.

Another potential problem arises when the manner in which alternatives are presented matters for choice. This is known as the *framing* problem. Consider the following example, paraphrased from Kahneman and Tversky (1984):

Imagine that you are about to purchase a stereo for 125 dollars and a calculator for 15 dollars. The salesman tells you that the calculator is on sale for 5 dollars less at the other branch of the store, located 20 minutes away. The stereo is the same price there. Would you make the trip to the other store?

It turns out that the fraction of respondents saying that they would travel to the other store for the 5 dollar discount is much higher than the fraction who say they would travel when the question is changed so that the 5 dollar saving is on the stereo. This is so even though the ultimate saving obtained by incurring the inconvenience of travel is the same in both

cases.³ Indeed, we would expect indifference to be the response to the following question:

Because of a stockout you must travel to the other store to get the two items, but you will receive 5 dollars off on either item as compensation. Do you care on which item this 5 dollar rebate is given?

If so, however, the individual violates transitivity. To see this, denote

- x = Travel to the other store and get a 5 dollar discount on the calculator.
- y = Travel to the other store and get a 5 dollar discount on the stereo.
- z = Buy both items at the first store.

The first two choices say that $x \succ z$ and $z \succ y$, but the last choice reveals $x \sim y$. Many problems of framing arise when individuals are faced with choices between alternatives that have uncertain outcomes (the subject of Chapter 6). Kahneman and Tversky (1984) provide a number of other interesting examples.

At the same time, it is often the case that apparently intransitive behavior can be explained fruitfully as the result of the interaction of several more primitive rational (and thus transitive) preferences. Consider the following two examples

(i) A household formed by Mom (M), Dad (D), and Child (C) makes decisions by majority voting. The alternatives for Friday evening entertainment are attending an opera (O), a rock concert (R), or an ice-skating show (I). The three members of the household have the rational individual preferences: $O \succ_M R \succ_M I$, $I \succ_D O \succ_D R$, $R \succ_C I \succ_C O$, where $\succ_M$, $\succ_D$, $\succ_C$ are the transitive individual strict preference relations. Now imagine three majority-rule votes: O versus R , R versus I , and I versus O . The result of these votes (O will win the first, R the second, and I the third) will make the household's preferences $\succ$ have the intransitive form: $O \succ R \succ I \succ O$. (The intransitivity illustrated in this example is known as the *Condorcet paradox*, and it is a central difficulty for the theory of group decision making. For further discussion, see Chapter 21.)

(ii) Intransitive decisions may also sometimes be viewed as a manifestation of a change of tastes. For example, a potential cigarette smoker may prefer smoking one cigarette a day to not smoking and may prefer not smoking to smoking heavily. But once she is smoking one cigarette a day, her tastes may change, and she may wish to increase the amount that she smokes. Formally, letting y be abstinence, x be smoking one cigarette a day, and z be heavy smoking, her initial situation is y , and her preferences in that initial situation are $x \succ y \succ z$. But once x is chosen over y and z , and there is a change of the individual's current situation from y to x , her tastes change to $z \succ x \succ y$. Thus, we apparently have an intransitivity: $z \succ x \succ z$. This *change-of-tastes* model has an important theoretical bearing on the analysis of addictive behavior. It also raises interesting issues related to commitment in decision making [see Schelling (1979)]. A rational decision maker will anticipate the induced change of tastes and will therefore attempt to tie her hand to her initial decision (Ulysses had himself tied to the mast when approaching the island of the Sirens).

It often happens that this change-of-tastes point of view gives us a well-structured way to think about *nonrational* decisions. See Elster (1979) for philosophical discussions of this and similar points.

Utility Functions

In economics, we often describe preference relations by means of a *utility function*. A utility function $u(x)$ assigns a numerical value to each element in X , ranking the

3. Kahneman and Tversky attribute this finding to individuals keeping "mental accounts" in which the savings are compared to the price of the item on which they are received.

elements of X in accordance with the individual's preferences. This is stated more precisely in Definition 1.B.2.

Definition 1.B.2: A function $u: X \rightarrow \mathbb{R}$ is a *utility function representing preference relation* $\succsim$ if, for all $x, y \in X$,

$$x \succsim y \Leftrightarrow u(x) \geq u(y).$$

Note that a utility function that represents a preference relation $\succsim$ is not unique. For any strictly increasing function $f: \mathbb{R} \rightarrow \mathbb{R}$, $v(x) = f(u(x))$ is a new utility function representing the same preferences as $u(\cdot)$; see Exercise 1.B.3. It is only the ranking of alternatives that matters. Properties of utility functions that are invariant for any strictly increasing transformation are called *ordinal*. *Cardinal* properties are those not preserved under all such transformations. Thus, the preference relation associated with a utility function is an ordinal property. On the other hand, the numerical values associated with the alternatives in X , and hence the magnitude of any differences in the utility measure between alternatives, are cardinal properties.

The ability to represent preferences by a utility function is closely linked to the assumption of rationality. In particular, we have the result shown in Proposition 1.B.2.

Proposition 1.B.2: A preference relation $\succsim$ can be represented by a utility function only if it is rational.

Proof: To prove this proposition, we show that if there is a utility function that represents preferences $\succsim$, then $\succsim$ must be complete and transitive.

Completeness. Because $u(\cdot)$ is a real-valued function defined on X , it must be that for any $x, y \in X$, either $u(x) \geq u(y)$ or $u(y) \geq u(x)$. But because $u(\cdot)$ is a utility function representing $\succsim$, this implies either that $x \succsim y$ or that $y \succsim x$ (recall Definition 1.B.2). Hence, $\succsim$ must be complete.

Transitivity. Suppose that $x \succsim y$ and $y \succsim z$. Because $u(\cdot)$ represents $\succsim$, we must have $u(x) \geq u(y)$ and $u(y) \geq u(z)$. Therefore, $u(x) \geq u(z)$. Because $u(\cdot)$ represents $\succsim$, this implies $x \succsim z$. Thus, we have shown that $x \succsim y$ and $y \succsim z$ imply $x \succsim z$, and so transitivity is established. ■

At the same time, one might wonder, can *any* rational preference relation $\succsim$ be described by some utility function? It turns out that, in general, the answer is no. An example where it is not possible to do so will be discussed in Section 3.G. One case in which we can always represent a rational preference relation with a utility function arises when X is finite (see Exercise 1.B.5). More interesting utility representation results (e.g., for sets of alternatives that are not finite) will be presented in later chapters.

1.C Choice Rules

In the second approach to the theory of decision making, choice behavior itself is taken to be the primitive object of the theory. Formally, choice behavior is represented by means of a *choice structure*. A choice structure $(\mathcal{B}, C(\cdot))$ consists of two ingredients:

(i) $\mathcal{B}$ is a family (a set) of nonempty subsets of X ; that is, every element of $\mathcal{B}$ is a set $B \subset X$. By analogy with the consumer theory to be developed in Chapters 2 and 3, we call the elements $B \in \mathcal{B}$ *budget sets*. The budget sets in $\mathcal{B}$ should be thought of as an exhaustive listing of all the choice experiments that the institutionally, physically, or otherwise restricted social situation can conceivably pose to the decision maker. It need not, however, include all possible subsets of X . Indeed, in the case of consumer demand studied in later chapters, it will not.

(ii) $C(\cdot)$ is a *choice rule* (technically, it is a correspondence) that assigns a nonempty set of chosen elements $C(B) \subset B$ for every budget set $B \in \mathcal{B}$. When $C(B)$ contains a single element, that element is the individual's choice from among the alternatives in B . The set $C(B)$ may, however, contain more than one element. When it does, the elements of $C(B)$ are the alternatives in B that the decision maker *might* choose; that is, they are her *acceptable alternatives* in B . In this case, the set $C(B)$ can be thought of as containing those alternatives that we would actually see chosen if the decision maker were repeatedly to face the problem of choosing an alternative from set B .

Example 1.C.1: Suppose that $X = \{x, y, z\}$ and $\mathcal{B} = \{\{x, y\}, \{x, y, z\}\}$. One possible choice structure is $(\mathcal{B}, C_1(\cdot))$, where the choice rule $C_1(\cdot)$ is: $C_1(\{x, y\}) = \{x\}$ and $C_1(\{x, y, z\}) = \{x\}$. In this case, we see x chosen no matter what budget the decision maker faces.

Another possible choice structure is $(\mathcal{B}, C_2(\cdot))$, where the choice rule $C_2(\cdot)$ is: $C_2(\{x, y\}) = \{x\}$ and $C_2(\{x, y, z\}) = \{x, y\}$. In this case, we see x chosen whenever the decision maker faces budget $\{x, y\}$, but we may see either x or y chosen when she faces budget $\{x, y, z\}$. ■

When using choice structures to model individual behavior, we may want to impose some “reasonable” restrictions regarding an individual's choice behavior. An important assumption, the weak axiom of revealed preference [first suggested by Samuelson; see Chapter 5 in Samuelson (1947)], reflects the expectation that an individual's observed choices will display a certain amount of consistency. For example, if an individual chooses alternative x (and only that) when faced with a choice between x and y , we would be surprised to see her choose y when faced with a decision among x , y , and a third alternative z . The idea is that the choice of x when facing the alternatives $\{x, y\}$ reveals a proclivity for choosing x over y that we should expect to see reflected in the individual's behavior when faced with the alternatives $\{x, y, z\}$.⁴

The weak axiom is stated formally in Definition 1.C.1.

Definition 1.C.1: The choice structure $(\mathcal{B}, C(\cdot))$ satisfies the *weak axiom of revealed preference* if the following property holds:

If for some $B \in \mathcal{B}$ with $x, y \in B$ we have $x \in C(B)$, then for any $B' \in \mathcal{B}$ with $x, y \in B'$ and $y \in C(B')$, we must also have $x \in C(B')$.

In words, the weak axiom says that if x is ever chosen when y is available, then there can be no budget set containing both alternatives for which y is chosen and x is not.

4. This proclivity might reflect some underlying “preference” for x over y but might also arise in other ways. It could, for example, be the result of some evolutionary process.

Note how the assumption that choice behavior satisfies the weak axiom captures the consistency idea: If $C(\{x, y\}) = \{x\}$, then the weak axiom says that we cannot have $C(\{x, y, z\}) = \{y\}$.⁵

A somewhat simpler statement of the weak axiom can be obtained by defining a *revealed preference relation* $\succsim^*$ from the observed choice behavior in $C(\cdot)$.

Definition 1.C.2: Given a choice structure $(\mathcal{B}, C(\cdot))$ the *revealed preference relation* $\succsim^*$ is defined by

$$x \succsim^* y \Leftrightarrow \text{there is some } B \in \mathcal{B} \text{ such that } x, y \in B \text{ and } x \in C(B).$$

We read $x \succsim^* y$ as “ x is revealed at least as good as y .” Note that the revealed preference relation $\succsim^*$ need not be either complete or transitive. In particular, for any pair of alternatives x and y to be comparable, it is necessary that, for some $B \in \mathcal{B}$, we have $x, y \in B$ and either $x \in C(B)$ or $y \in C(B)$, or both.

We might also informally say that “ x is revealed preferred to y ” if there is some $B \in \mathcal{B}$ such that $x, y \in B$, $x \in C(B)$, and $y \notin C(B)$, that is, if x is ever chosen over y when both are feasible.

With this terminology, we can restate the weak axiom as follows: “If x is revealed at least as good as y , then y cannot be revealed preferred to x .”

Example 1.C.2: Do the two choice structures considered in Example 1.C.1 satisfy the weak axiom? Consider choice structure $(\mathcal{B}, C_1(\cdot))$. With this choice structure, we have $x \succsim^* y$ and $x \succsim^* z$, but there is no revealed preference relationship that can be inferred between y and z . This choice structure satisfies the weak axiom because y and z are never chosen.

Now consider choice structure $(\mathcal{B}, C_2(\cdot))$. Because $C_2(\{x, y, z\}) = \{x, y\}$, we have $y \succsim^* x$ (as well as $x \succsim^* y$, $x \succsim^* z$, and $y \succsim^* z$). But because $C_2(\{x, y\}) = \{x\}$, x is revealed preferred to y . Therefore, the choice structure $(\mathcal{B}, C_2)$ violates the weak axiom. ■

We should note that the weak axiom is not the only assumption concerning choice behavior that we may want to impose in any particular setting. For example, in the consumer demand setting discussed in Chapter 2, we impose further conditions that arise naturally in that context.

The weak axiom restricts choice behavior in a manner that parallels the use of the rationality assumption for preference relations. This raises a question: What is the precise relationship between the two approaches? In Section 1.D, we explore this matter.

1.D The Relationship between Preference Relations and Choice Rules

We now address two fundamental questions regarding the relationship between the two approaches discussed so far:

5. In fact, it says more: We must have $C(\{x, y, z\}) = \{x\}$, $= \{z\}$, or $= \{x, z\}$. You are asked to show this in Exercise 1.C.1. See also Exercise 1.C.2.

- (i) If a decision maker has a rational preference ordering $\succsim$, do her decisions when facing choices from budget sets in $\mathcal{B}$ necessarily generate a choice structure that satisfies the weak axiom?
- (ii) If an individual's choice behavior for a family of budget sets $\mathcal{B}$ is captured by a choice structure $(\mathcal{B}, C(\cdot))$ satisfying the weak axiom, is there necessarily a rational preference relation that is consistent with these choices?

As we shall see, the answers to these two questions are, respectively, “yes” and “maybe”.

To answer the first question, suppose that an individual has a rational preference relation $\succsim$ on X . If this individual faces a nonempty subset of alternatives $B \subset X$, her preference-maximizing behavior is to choose any one of the elements in the set:

$$C^*(B, \succsim) = \{x \in B: x \succsim y \text{ for every } y \in B\}$$

The elements of set $C^*(B, \succsim)$ are the decision maker's most preferred alternatives in B . In principle, we could have $C^*(B, \succsim) = \emptyset$ for some B ; but if X is finite, or if suitable (continuity) conditions hold, then $C^*(B, \succsim)$ will be nonempty.⁶ From now on, we will consider only preferences $\succsim$ and families of budget sets $\mathcal{B}$ such that $C^*(B, \succsim)$ is nonempty for all $B \in \mathcal{B}$. We say that the rational preference relation $\succsim$ *generates* the choice structure $(\mathcal{B}, C^*(\cdot, \succsim))$.

The result in Proposition 1.D.1 tells us that any choice structure generated by rational preferences necessarily satisfies the weak axiom.

Proposition 1.D.1: Suppose that $\succsim$ is a rational preference relation. Then the choice structure generated by $\succsim$, $(\mathcal{B}, C^*(\cdot, \succsim))$, satisfies the weak axiom.

Proof: Suppose that for some $B \in \mathcal{B}$, we have $x, y \in B$ and $x \in C^*(B, \succsim)$. By the definition of $C^*(B, \succsim)$, this implies $x \succsim y$. To check whether the weak axiom holds, suppose that for some $B' \in \mathcal{B}$ with $x, y \in B'$, we have $y \in C^*(B', \succsim)$. This implies that $y \succsim z$ for all $z \in B'$. But we already know that $x \succsim y$. Hence, by transitivity, $x \succsim z$ for all $z \in B'$, and so $x \in C^*(B', \succsim)$. This is precisely the conclusion that the weak axiom demands. ■

Proposition 1.D.1 constitutes the “yes” answer to our first question. That is, if behavior is generated by rational preferences then it satisfies the consistency requirements embodied in the weak axiom.

In the other direction (from choice to preferences), the relationship is more subtle. To answer this second question, it is useful to begin with a definition.

Definition 1.D.1: Given a choice structure $(\mathcal{B}, C(\cdot))$, we say that the rational preference relation $\succsim$ *rationalizes* $C(\cdot)$ relative to $\mathcal{B}$ if

$$C(B) = C^*(B, \succsim)$$

for all $B \in \mathcal{B}$, that is, if $\succsim$ generates the choice structure $(\mathcal{B}, C(\cdot))$.

In words, the rational preference relation $\succsim$ rationalizes choice rule $C(\cdot)$ on $\mathcal{B}$ if the optimal choices generated by $\succsim$ (captured by $C^*(\cdot, \succsim)$) coincide with $C(\cdot)$ for

6. Exercise 1.D.2 asks you to establish the nonemptiness of $C^*(B, \succsim)$ for the case where X is finite. For general results, See Section M.F of the Mathematical Appendix and Section 3.C for a specific application.

all budget sets in $\mathcal{B}$. In a sense, preferences explain behavior; we can interpret the decision maker's choices as if she were a preference maximizer. Note that in general, there may be more than one rationalizing preference relation $\succsim$ for a given choice structure $(\mathcal{B}, C(\cdot))$ (see Exercise 1.D.1).

Proposition 1.D.1 implies that the weak axiom must be satisfied if there is to be a rationalizing preference relation. In particular, since $C^*(\cdot, \succsim)$ satisfies the weak axiom for any $\succsim$, only a choice rule that satisfies the weak axiom can be rationalized. It turns out, however, that the weak axiom is not sufficient to ensure the existence of a rationalizing preference relation.

Example 1.D.1: Suppose that $X = \{x, y, z\}$, $\mathcal{B} = \{\{x, y\}, \{y, z\}, \{x, z\}\}$, $C(\{x, y\}) = \{x\}$, $C(\{y, z\}) = \{y\}$, and $C(\{x, z\}) = \{z\}$. This choice structure satisfies the weak axiom (you should verify this). Nevertheless, we cannot have rationalizing preferences. To see this, note that to rationalize the choices under $\{x, y\}$ and $\{y, z\}$ it would be necessary for us to have $x \succ y$ and $y \succ z$. But, by transitivity, we would then have $x \succ z$, which contradicts the choice behavior under $\{x, z\}$. Therefore, there can be no rationalizing preference relation. ■

To understand Example 1.D.1, note that the more budget sets there are in $\mathcal{B}$, the more the weak axiom restricts choice behavior; there are simply more opportunities for the decision maker's choices to contradict one another. In Example 1.D.1, the set $\{x, y, z\}$ is not an element of $\mathcal{B}$. As it happens, this is crucial (see Exercises 1.D.3). As we now show in Proposition 1.D.2, if the family of budget sets $\mathcal{B}$ includes enough subsets of X , and if $(\mathcal{B}, C(\cdot))$ satisfies the weak axiom, then there exists a rational preference relation that rationalizes $C(\cdot)$ relative to $\mathcal{B}$ [this was first shown by Arrow (1959)].

Proposition 1.D.2: If $(\mathcal{B}, C(\cdot))$ is a choice structure such that

- (i) the weak axiom is satisfied,
- (ii) $\mathcal{B}$ includes all subsets of X of up to three elements,

then there is a rational preference relation $\succsim$ that rationalizes $C(\cdot)$ relative to $\mathcal{B}$; that is, $C(B) = C^*(B, \succsim)$ for all $B \in \mathcal{B}$. Furthermore, this rational preference relation is the *only* preference relation that does so.

Proof: The natural candidate for a rationalizing preference relation is the revealed preference relation $\succsim^*$. To prove the result, we must first show two things: (i) that $\succsim^*$ is a rational preference relation, and (ii) that $\succsim^*$ rationalizes $C(\cdot)$ on $\mathcal{B}$. We then argue, as point (iii), that $\succsim^*$ is the unique preference relation that does so.

- (i) We first check that $\succsim^*$ is rational (i.e., that it satisfies completeness and transitivity).

Completeness By assumption (ii), $\{x, y\} \in \mathcal{B}$. Since either x or y must be an element of $C(\{x, y\})$, we must have $x \succsim^* y$, or $y \succsim^* x$, or both. Hence $\succsim^*$ is complete.

Transitivity Let $x \succsim^* y$ and $y \succsim^* z$. Consider the budget set $\{x, y, z\} \in \mathcal{B}$. It suffices to prove that $x \in C(\{x, y, z\})$, since this implies by the definition of $\succsim^*$ that $x \succsim^* z$. Because $C(\{x, y, z\}) \neq \emptyset$, at least one of the alternatives x, y , or z must be an element of $C(\{x, y, z\})$. Suppose that $y \in C(\{x, y, z\})$. Since $x \succsim^* y$, the weak axiom then yields $x \in C(\{x, y, z\})$, as we want. Suppose instead that $z \in C(\{x, y, z\})$; since $y \succsim^* z$, the weak axiom yields $y \in C(\{x, y, z\})$, and we are in the previous case.

- (ii) We now show that $C(B) = C^*(B, \succsim^*)$ for all $B \in \mathcal{B}$; that is, the revealed preference

relation $\succsim^*$ inferred from $C(\cdot)$ actually generates $C(\cdot)$. Intuitively, this seems sensible. Formally, we show this in two steps. First, suppose that $x \in C(B)$. Then $x \succsim^* y$ for all $y \in B$; so we have $x \in C^*(B, \succsim^*)$. This means that $C(B) \subset C^*(B, \succsim^*)$. Next, suppose that $x \in C^*(B, \succsim^*)$. This implies that $x \succsim^* y$ for all $y \in B$; and so for each $y \in B$, there must exist some set $B_y \in \mathcal{B}$ such that $x, y \in B_y$ and $x \in C(B_y)$. Because $C(B) \neq \emptyset$, the weak axiom then implies that $x \in C(B)$. Hence, $C^*(B, \succsim^*) \subset C(B)$. Together, these inclusion relations imply that $C(B) = C^*(B, \succsim^*)$.

(iii) To establish uniqueness, simply note that because $\mathcal{B}$ includes all two-element subsets of X , the choice behavior in $C(\cdot)$ completely determines the pairwise preference relations over X of any rationalizing preference.

This completes the proof. ■

We can therefore conclude from Proposition 1.D.2 that for the special case in which choice is defined for all subsets of X , a theory based on choice satisfying the weak axiom is completely equivalent to a theory of decision making based on rational preferences. Unfortunately, this special case is too special for economics. For many situations of economic interest, such as the theory of consumer demand, choice is defined only for special kinds of budget sets. In these settings, the weak axiom does not exhaust the choice implications of rational preferences. We shall see in Section 3.J, however, that a strengthening of the weak axiom (which imposes more restrictions on choice behavior) provides a necessary and sufficient condition for behavior to be capable of being rationalized by preferences.

Definition 1.D.1 defines a rationalizing preference as one for which $C(B) = C^*(B, \succsim)$. An alternative notion of a rationalizing preference that appears in the literature requires only that $C(B) \subset C^*(B, \succsim)$; that is, $\succsim$ is said to rationalize $C(\cdot)$ on $\mathcal{B}$ if $C(B)$ is a subset of the most preferred choices generated by $\succsim$, $C^*(B, \succsim)$, for every budget $B \in \mathcal{B}$.

There are two reasons for the possible use of this alternative notion. The first is, in a sense, philosophical. We might want to allow the decision maker to resolve her indifference in some specific manner, rather than insisting that indifference means that anything might be picked. The view embodied in Definition 1.D.1 (and implicitly in the weak axiom as well) is that if she chooses in a specific manner then she is, *de facto*, not indifferent.

The second reason is empirical. If we are trying to determine from data whether an individual's choice is compatible with rational preference maximization, we will in practice have only a finite number of observations on the choices made from any given budget set B . If $C(B)$ represents the set of choices made with this limited set of observations, then because these limited observations might not reveal all the decision maker's preference maximizing choices, $C(B) \subset C^*(B, \succsim)$ is the natural requirement to impose for a preference relationship to rationalize observed choice data.

Two points are worth noting about the effects of using this alternative notion. First, it is a weaker requirement. Whenever we can find a preference relation that rationalizes choice in the sense of Definition 1.D.1, we have found one that does so in this other sense, too. Second, in the abstract setting studied here, to find a rationalizing preference relation in this latter sense is actually trivial: Preferences that have the individual indifferent among all elements of X will rationalize *any* choice behavior in this sense. When this alternative notion is used in the economics literature, there is always an insistence that the rationalizing preference relation should satisfy some additional properties that are natural restrictions for the specific economic context being studied.

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EXERCISES

1.B.1^B Prove property (iii) of Proposition 1.B.1.

1.B.2^A Prove properties (i) and (ii) of Proposition 1.B.1.

1.B.3^B Show that if $f: \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing function and $u: X \rightarrow \mathbb{R}$ is a utility function representing preference relation $\succsim$, then the function $v: X \rightarrow \mathbb{R}$ defined by $v(x) = f(u(x))$ is also a utility function representing preference relation $\succsim$.

1.B.4^A Consider a rational preference relation $\succsim$. Show that if $u(x) = u(y)$ implies $x \sim y$ and if $u(x) > u(y)$ implies $x \succ y$, then $u(\cdot)$ is a utility function representing $\succsim$.

1.B.5^B Show that if X is finite and $\succsim$ is a rational preference relation on X , then there is a utility function $u: X \rightarrow \mathbb{R}$ that represents $\succsim$. [Hint: Consider first the case in which the individual's ranking between any two elements of X is strict (i.e., there is never any indifference), and construct a utility function representing these preferences; then extend your argument to the general case.]

1.C.1^B Consider the choice structure $(\mathcal{B}, C(\cdot))$ with $\mathcal{B} = (\{x, y\}, \{x, y, z\})$ and $C(\{x, y\}) = \{x\}$. Show that if $(\mathcal{B}, C(\cdot))$ satisfies the weak axiom, then we must have $C(\{x, y, z\}) = \{x\}$, $\{z\}$, or $\{x, z\}$.

1.C.2^B Show that the weak axiom (Definition 1.C.1) is equivalent to the following property holding:

Suppose that $B, B' \in \mathcal{B}$, that $x, y \in B$, and that $x, y \in B'$. Then if $x \in C(B)$ and $y \in C(B')$, we must have $\{x, y\} \subset C(B)$ and $\{x, y\} \subset C(B')$.

1.C.3^C Suppose that choice structure $(\mathcal{B}, C(\cdot))$ satisfies the weak axiom. Consider the following two possible revealed preferred relations, $\succ^*$ and $\succ^{**}$:

$$x \succ^* y \Leftrightarrow \text{there is some } B \in \mathcal{B} \text{ such that } x, y \in B, x \in C(B), \text{ and } y \notin C(B)$$

$$x \succ^{**} y \Leftrightarrow x \succ^* y \text{ but not } y \succ^* x$$

where $\succ^*$ is the revealed at-least-as-good-as relation defined in Definition 1.C.2.

(a) Show that $\succ^*$ and $\succ^{**}$ give the same relation over X ; that is, for any $x, y \in X$, $x \succ^* y \Leftrightarrow x \succ^{**} y$. Is this still true if $(\mathcal{B}, C(\cdot))$ does not satisfy the weak axiom?

(b) Must $\succ^*$ be transitive?

(c) Show that if $\mathcal{B}$ includes all three-element subsets of X , then $\succ^*$ is transitive.

1.D.1^B Give an example of a choice structure that can be rationalized by several preference relations. Note that if the family of budgets $\mathcal{B}$ includes all the two-element subsets of X , then there can be at most one rationalizing preference relation.

1.D.2^A Show that if X is finite, then any rational preference relation generates a nonempty choice rule; that is, $C(B) \neq \emptyset$ for any $B \subset X$ with $B \neq \emptyset$.

1.D.3^B Let $X = \{x, y, z\}$, and consider the choice structure $(\mathcal{B}, C(\cdot))$ with

$$\mathcal{B} = \{\{x, y\}, \{y, z\}, \{x, z\}, \{x, y, z\}\}$$

and $C(\{x, y\}) = \{x\}$, $C(\{y, z\}) = \{y\}$, and $C(\{x, z\}) = \{z\}$, as in Example 1.D.1. Show that $(\mathcal{B}, C(\cdot))$ must violate the weak axiom.

1.D.4^B Show that a choice structure $(\mathcal{B}, C(\cdot))$ for which a rationalizing preference relation $\succsim$ exists satisfies the *path-invariance* property: For every pair $B_1, B_2 \in \mathcal{B}$ such that $B_1 \cup B_2 \in \mathcal{B}$ and $C(B_1) \cup C(B_2) \in \mathcal{B}$, we have $C(B_1 \cup B_2) = C(C(B_1) \cup C(B_2))$, that is, the decision problem can safely be subdivided. See Plott (1973) for further discussion.

1.D.5^C Let $X = \{x, y, z\}$ and $\mathcal{B} = \{\{x, y\}, \{y, z\}, \{z, x\}\}$. Suppose that choice is now stochastic in the sense that, for every $B \in \mathcal{B}$, $C(B)$ is a frequency distribution over alternatives in B . For example, if $B = \{x, y\}$, we write $C(B) = (C_x(B), C_y(B))$, where $C_x(B)$ and $C_y(B)$ are nonnegative numbers with $C_x(B) + C_y(B) = 1$. We say that the stochastic choice function $C(\cdot)$ can be *rationalized by preferences* if we can find a probability distribution Pr over the six possible (strict) preference relations on X such that for every $B \in \mathcal{B}$, $C(B)$ is precisely the frequency of choices induced by Pr . For example, if $B = \{x, y\}$, then $C_x(B) = Pr(\{>: x > y\})$. This concept originates in Thurstone (1927), and it is of considerable econometric interest (indeed, it provides a theory for the error term in observable choice).

- (a) Show that the stochastic choice function $C(\{x, y\}) = C(\{y, z\}) = C(\{z, x\}) = (\frac{1}{2}, \frac{1}{2})$ can be rationalized by preferences.
- (b) Show that the stochastic choice function $C(\{x, y\}) = C(\{y, z\}) = C(\{z, x\}) = (\frac{1}{4}, \frac{3}{4})$ is not rationalizable by preferences.
- (c) Determine the $0 < \alpha < 1$ at which $C(\{x, y\}) = C(\{y, z\}) = C(\{z, x\}) = (\alpha, 1 - \alpha)$ switches from rationalizable to nonrationalizable.

Consumer Choice

2.A Introduction

The most fundamental decision unit of microeconomic theory is the *consumer*. In this chapter, we begin our study of consumer demand in the context of a market economy. By a *market economy*, we mean a setting in which the goods and services that the consumer may acquire are available for purchase at known prices (or, equivalently, are available for trade for other goods at known rates of exchange).

We begin, in Sections 2.B to 2.D, by describing the basic elements of the consumer's decision problem. In Section 2.B, we introduce the concept of *commodities*, the objects of choice for the consumer. Then, in Sections 2.C and 2.D, we consider the physical and economic constraints that limit the consumer's choices. The former are captured in the *consumption set*, which we discuss in Section 2.C; the latter are incorporated in Section 2.D into the consumer's *Walrasian budget set*.

The consumer's decision subject to these constraints is captured in the consumer's *Walrasian demand function*. In terms of the choice-based approach to individual decision making introduced in Section 1.C, the Walrasian demand function is the consumer's choice rule. We study this function and some of its basic properties in Section 2.E. Among them are what we call *comparative statics* properties: the ways in which consumer demand changes when economic constraints vary.

Finally, in Section 2.F, we consider the implications for the consumer's demand function of the *weak axiom of revealed preference*. The central conclusion we reach is that in the consumer demand setting, the weak axiom is essentially equivalent to the *compensated law of demand*, the postulate that prices and demanded quantities move in opposite directions for price changes that leave real wealth unchanged.

2.B Commodities

The decision problem faced by the consumer in a market economy is to choose consumption levels of the various goods and services that are available for purchase in the market. We call these goods and services *commodities*. For simplicity, we assume that the number of commodities is finite and equal to L (indexed by $\ell = 1, \dots, L$).

As a general matter, a *commodity vector* (or *commodity bundle*) is a list of amounts of the different commodities,

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_L \end{bmatrix},$$

and can be viewed as a point in $\mathbb{R}^L$, the commodity *space*.¹

We can use commodity vectors to represent an individual's consumption levels. The ℓ th entry of the commodity vector stands for the amount of commodity ℓ consumed. We then refer to the vector as a *consumption vector* or *consumption bundle*.

Note that time (or, for that matter, location) can be built into the definition of a commodity. Rigorously, bread today and tomorrow should be viewed as distinct commodities. In a similar vein, when we deal with decisions under uncertainty in Chapter 6, viewing bread in different “states of nature” as different commodities can be most helpful.

Although commodities consumed at different times should be viewed rigorously as distinct commodities, in practice, economic models often involve some “time aggregation.” Thus, one commodity might be “bread consumed in the month of February,” even though, in principle, bread consumed at each instant in February should be distinguished. A primary reason for such time aggregation is that the economic data to which the model is being applied are aggregated in this way. The hope of the modeler is that the commodities being aggregated are sufficiently similar that little of economic interest is being lost.

We should also note that in some contexts it becomes convenient, and even necessary, to expand the set of commodities to include goods and services that may potentially be available for purchase but are not actually so and even some that may be available by means other than market exchange (say, the experience of “family togetherness”). For nearly all of what follows here, however, the narrow construction introduced in this section suffices.

2.C The Consumption Set

Consumption choices are typically limited by a number of physical constraints. The simplest example is when it may be impossible for the individual to consume a negative amount of a commodity such as bread or water.

Formally, the *consumption set* is a subset of the commodity space $\mathbb{R}^L$, denoted by $X \subset \mathbb{R}^L$, whose elements are the consumption bundles that the individual can conceivably consume given the physical constraints imposed by his environment.

Consider the following four examples for the case in which $L = 2$:

- (i) Figure 2.C.1 represents possible consumption levels of bread and leisure in a day. Both levels must be nonnegative and, in addition, the consumption of more than 24 hours of leisure in a day is impossible.
- (ii) Figure 2.C.2 represents a situation in which the first good is perfectly divisible but the second is available only in nonnegative integer amounts.
- (iii) Figure 2.C.3 captures the fact that it is impossible to eat bread at the same

1. Negative entries in commodity vectors will often represent debits or net outflows of goods. For example, in Chapter 5, the inputs of a firm are measured as negative numbers.

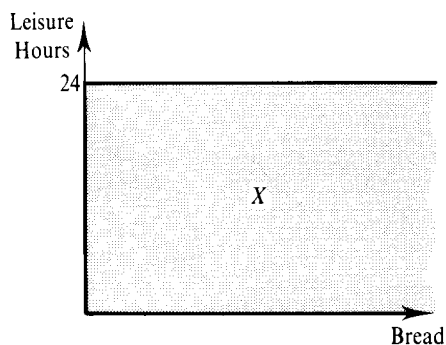


Figure 2.C.1 (left)
A consumption set.

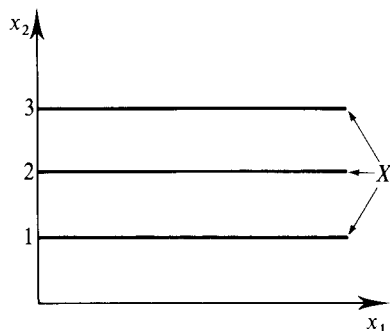


Figure 2.C.2 (right)
A consumption set where good 2 must be consumed in integer amounts.

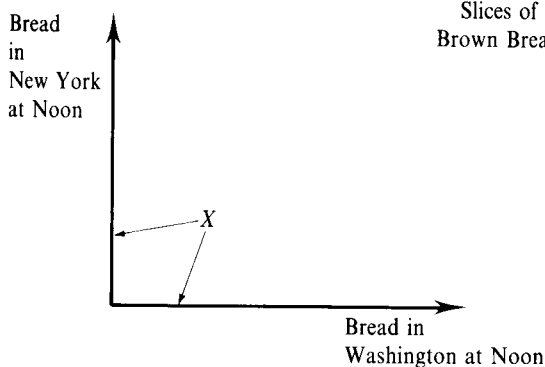


Figure 2.C.3 (left)
A consumption set where only one good can be consumed.

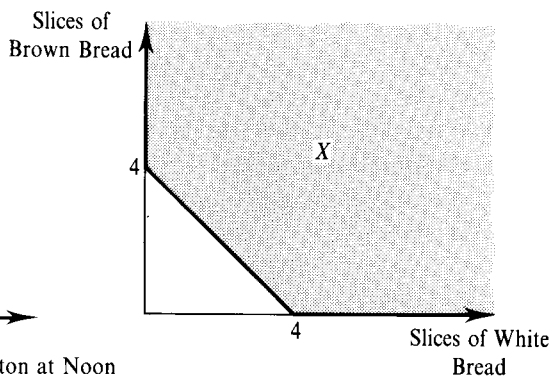


Figure 2.C.4 (right)
A consumption set reflecting survival needs.

instant in Washington and in New York. [This example is borrowed from Malinvaud (1978).]

- (iv) Figure 2.C.4 represents a situation where the consumer requires a minimum of four slices of bread a day to survive and there are two types of bread, brown and white.

In the four examples, the constraints are physical in a very literal sense. But the constraints that we incorporate into the consumption set can also be institutional in nature. For example, a law requiring that no one work more than 16 hours a day would change the consumption set in Figure 2.C.1 to that in Figure 2.C.5.

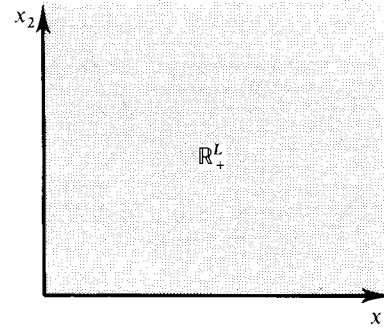
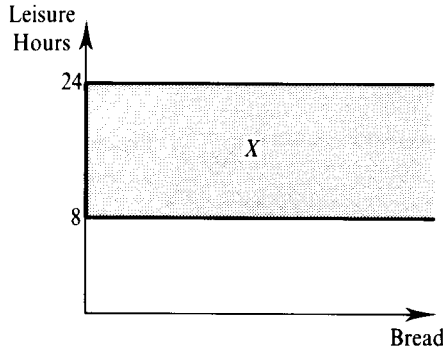
To keep things as straightforward as possible, we pursue our discussion adopting the simplest sort of consumption set:

$$X = \mathbb{R}_+^L = \{x \in \mathbb{R}^L: x_\ell \geq 0 \text{ for } \ell = 1, \dots, L\},$$

the set of all nonnegative bundles of commodities. It is represented in Figure 2.C.6. Whenever we consider any consumption set X other than $\mathbb{R}_+^L$, we shall be explicit about it.

One special feature of the set $\mathbb{R}_+^L$ is that it is *convex*. That is, if two consumption bundles x and x' are both elements of $\mathbb{R}_+^L$, then the bundle $x'' = \alpha x + (1 - \alpha)x'$ is also an element of $\mathbb{R}_+^L$ for any $\alpha \in [0, 1]$ (see Section M.G. of the Mathematical Appendix for the definition and properties of convex sets).² The consumption sets

2. Recall that $x'' = \alpha x + (1 - \alpha)x'$ is a vector whose ℓ th entry is $x''_\ell = \alpha x_\ell + (1 - \alpha)x'_\ell$.

**Figure 2.C.5 (left)**

A consumption set reflecting a legal limit on the number of hours worked.

Figure 2.C.6 (right)

The consumption set $\mathbb{R}_+^L$.

in Figures 2.C.1, 2.C.4, 2.C.5, and 2.C.6 are convex sets; those in Figures 2.C.2 and 2.C.3 are not.

Much of the theory to be developed applies for general convex consumption sets as well as for $\mathbb{R}_+^L$. Some of the results, but not all, survive without the assumption of convexity.³

2.D Competitive Budgets

In addition to the physical constraints embodied in the consumption set, the consumer faces an important economic constraint: his consumption choice is limited to those commodity bundles that he can afford.

To formalize this constraint, we introduce two assumptions. First, we suppose that the L commodities are all traded in the market at dollar prices that are publicly quoted (this is the *principle of completeness*, or *universality, of markets*). Formally, these prices are represented by the *price vector*

$$p = \begin{bmatrix} p_1 \\ \vdots \\ p_L \end{bmatrix} \in \mathbb{R}^L,$$

which gives the dollar cost for a unit of each of the L commodities. Observe that there is nothing that logically requires prices to be positive. A negative price simply means that a “buyer” is actually paid to consume the commodity (which is not illogical for commodities that are “bads,” such as pollution). Nevertheless, for simplicity, here we always assume $p \gg 0$; that is, $p_\ell > 0$ for every ℓ .

Second, we assume that these prices are beyond the influence of the consumer. This is the so-called *price-taking assumption*. Loosely speaking, this assumption is likely to be valid when the consumer’s demand for any commodity represents only a small fraction of the total demand for that good.

The affordability of a consumption bundle depends on two things: the market prices $p = (p_1, \dots, p_L)$ and the consumer’s wealth level (in dollars) w . The consumption

3. Note that commodity aggregation can help convexify the consumption set. In the example leading to Figure 2.C.3, the consumption set could reasonably be taken to be convex if the axes were instead measuring bread consumption over a period of a month.

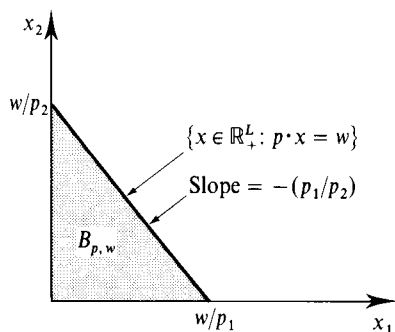


Figure 2.D.1 (left)
A Walrasian budget set.

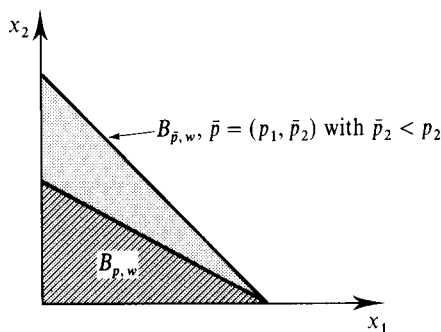


Figure 2.D.2 (right)
The effect of a price change on the Walrasian budget set.

bundle $x \in \mathbb{R}_+^L$ is affordable if its total cost does not exceed the consumer's wealth level w , that is, if⁴

$$p \cdot x = p_1 x_1 + \cdots + p_L x_L \leq w.$$

This economic-affordability constraint, when combined with the requirement that x lie in the consumption set $\mathbb{R}_+^L$, implies that the set of feasible consumption bundles consists of the elements of the set $\{x \in \mathbb{R}_+^L : p \cdot x \leq w\}$. This set is known as the *Walrasian*, or *competitive budget set* (after Léon Walras).

Definition 2.D.1: The *Walrasian, or competitive budget set* $B_{p,w} = \{x \in \mathbb{R}_+^L : p \cdot x \leq w\}$ is the set of all feasible consumption bundles for the consumer who faces market prices p and has wealth w .

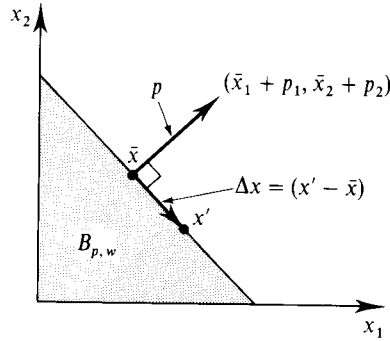
The *consumer's problem*, given prices p and wealth w , can thus be stated as follows: Choose a consumption bundle x from $B_{p,w}$.

A Walrasian budget set $B_{p,w}$ is depicted in Figure 2.D.1 for the case of $L = 2$. To focus on the case in which the consumer has a nondegenerate choice problem, we always assume $w > 0$ (otherwise the consumer can afford only $x = 0$).

The set $\{x \in \mathbb{R}_+^L : p \cdot x = w\}$ is called the *budget hyperplane* (for the case $L = 2$, we call it the *budget line*). It determines the upper boundary of the budget set. As Figure 2.D.1 indicates, the slope of the budget line when $L = 2$, $-(p_1/p_2)$, captures the rate of exchange between the two commodities. If the price of commodity 2 decreases (with p_1 and w held fixed), say to $\bar{p}_2 < p_2$, the budget set grows larger because more consumption bundles are affordable, and the budget line becomes steeper. This change is shown in Figure 2.D.2.

Another way to see how the budget hyperplane reflects the relative terms of exchange between commodities comes from examining its geometric relation to the price vector p . The price vector p , drawn starting from any point $\bar{x}$ on the budget hyperplane, must be orthogonal (perpendicular) to any vector starting at $\bar{x}$ and lying

4. Often, this constraint is described in the literature as requiring that the cost of planned purchases not exceed the consumer's *income*. In either case, the idea is that the cost of purchases not exceed the consumer's available resources. We use the wealth terminology to emphasize that the consumer's actual problem may be intertemporal, with the commodities involving purchases over time, and the resource constraint being one of lifetime income (i.e., wealth) (see Exercise 2.D.1).

**Figure 2.D.3**

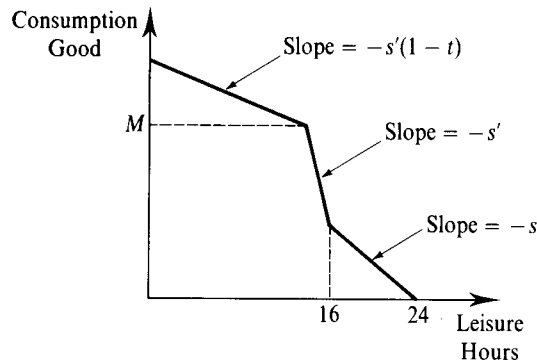
The geometric relationship between p and the budget hyperplane.

on the budget hyperplane. This is so because for any x' that itself lies on the budget hyperplane, we have $p \cdot x' = p \cdot \bar{x} = w$. Hence, $p \cdot \Delta x = 0$ for $\Delta x = (x' - \bar{x})$. Figure 2.D.3 depicts this geometric relationship for the case $L = 2$.⁵

The Walrasian budget set $B_{p,w}$ is a *convex* set: That is, if bundles x and x' are both elements of $B_{p,w}$, then the bundle $x'' = \alpha x + (1 - \alpha)x'$ is also. To see this, note first that because both x and x' are nonnegative, $x'' \in \mathbb{R}_+^L$. Second, since $p \cdot x \leq w$ and $p \cdot x' \leq w$, we have $p \cdot x'' = \alpha(p \cdot x) + (1 - \alpha)(p \cdot x') \leq w$. Thus, $x'' \in B_{p,w} = \{x \in \mathbb{R}_+^L : p \cdot x \leq w\}$.

The convexity of $B_{p,w}$ plays a significant role in the development that follows. Note that the convexity of $B_{p,w}$ depends on the convexity of the consumption set $\mathbb{R}_+^L$. With a more general consumption set X , $B_{p,w}$ will be convex as long as X is. (See Exercise 2.D.3.)

Although Walrasian budget sets are of central theoretical interest, they are by no means the only type of budget set that a consumer might face in any actual situation. For example, a more realistic description of the market trade-off between a consumption good and leisure, involving taxes, subsidies, and several wage rates, is illustrated in Figure 2.D.4. In the figure, the price of the consumption good is 1, and the consumer earns wage rate s per hour for the first 8 hours of work and $s' > s$ for additional (“overtime”) hours. He also faces a tax rate t

**Figure 2.D.4**

A more realistic description of the consumer's budget set.

5. To draw the vector p starting from $\bar{x}$, we draw a vector from point $(\bar{x}_1, \bar{x}_2)$ to point $(\bar{x}_1 + p_1, \bar{x}_2 + p_2)$. Thus, when we draw the price vector in this diagram, we use the “units” on the axes to represent units of prices rather than goods.

per dollar on labor income earned above amount M . Note that the budget set in Figure 2.D.4 is not convex (you are asked to show this in Exercise 2.D.4). More complicated examples can readily be constructed and arise commonly in applied work. See Deaton and Muellbauer (1980) and Burtless and Hausmann (1975) for more illustrations of this sort.

2.E Demand Functions and Comparative Statics

The consumer's *Walrasian* (or *market*, or *ordinary*) *demand correspondence* $x(p, w)$ assigns a set of chosen consumption bundles for each price–wealth pair (p, w) . In principle, this correspondence can be multivalued; that is, there may be more than one possible consumption vector assigned for a given price–wealth pair (p, w) . When this is so, any $x \in x(p, w)$ might be chosen by the consumer when he faces price–wealth pair (p, w) . When $x(p, w)$ is single-valued, we refer to it as a *demand function*.

Throughout this chapter, we maintain two assumptions regarding the Walrasian demand correspondence $x(p, w)$: That it is *homogeneous of degree zero* and that it satisfies *Walras' law*.

Definition 2.E.1: The Walrasian demand correspondence $x(p, w)$ is *homogeneous of degree zero* if $x(\alpha p, \alpha w) = x(p, w)$ for any p, w and $\alpha > 0$.

Homogeneity of degree zero says that if both prices and wealth change in the same proportion, then the individual's consumption choice does not change. To understand this property, note that a change in prices and wealth from (p, w) to $(\alpha p, \alpha w)$ leads to no change in the consumer's set of feasible consumption bundles; that is, $B_{p,w} = B_{\alpha p, \alpha w}$. Homogeneity of degree zero says that the individual's choice depends only on the set of feasible points.

Definition 2.E.2: The Walrasian demand correspondence $x(p, w)$ satisfies *Walras' law* if for every $p \gg 0$ and $w > 0$, we have $p \cdot x = w$ for all $x \in x(p, w)$.

Walras' law says that the consumer fully expends his wealth. Intuitively, this is a reasonable assumption to make as long as there is some good that is clearly desirable. Walras' law should be understood broadly: the consumer's budget may be an intertemporal one allowing for savings today to be used for purchases tomorrow. What Walras' law says is that the consumer fully expends his resources *over his lifetime*.

Exercise 2.E.1: Suppose $L = 3$, and consider the demand function $x(p, w)$ defined by

$$\begin{aligned} x_1(p, w) &= \frac{p_2}{p_1 + p_2 + p_3} \frac{w}{p_1}, \\ x_2(p, w) &= \frac{p_3}{p_1 + p_2 + p_3} \frac{w}{p_2}, \\ x_3(p, w) &= \frac{\beta p_1}{p_1 + p_2 + p_3} \frac{w}{p_3}. \end{aligned}$$

Does this demand function satisfy homogeneity of degree zero and Walras' law when $\beta = 1$? What about when $\beta \in (0, 1)$?

In Chapter 3, where the consumer's demand $x(p, w)$ is derived from the maximization of preferences, these two properties (homogeneity of degree zero and satisfaction of Walras' law) hold under very general circumstances. In the rest of this chapter, however, we shall simply take them as assumptions about $x(p, w)$ and explore their consequences.

One convenient implication of $x(p, w)$ being homogeneous of degree zero can be noted immediately: Although $x(p, w)$ formally has $L + 1$ arguments, we can, with no loss of generality, fix (*normalize*) the level of one of the $L + 1$ independent variables at an arbitrary level. One common normalization is $p_\ell = 1$ for some ℓ . Another is $w = 1$.⁶ Hence, the effective number of arguments in $x(p, w)$ is L .

For the remainder of this section, we assume that $x(p, w)$ is always single-valued. In this case, we can write the function $x(p, w)$ in terms of commodity-specific demand functions:

$$x(p, w) = \begin{bmatrix} x_1(p, w) \\ x_2(p, w) \\ \vdots \\ x_L(p, w) \end{bmatrix}.$$

When convenient, we also assume $x(p, w)$ to be continuous and differentiable.

The approach we take here and in Section 2.F can be viewed as an application of the choice-based framework developed in Chapter 1. The family of Walrasian budget sets is $\mathcal{B}^w = \{B_{p,w}: p \gg 0, w > 0\}$. Moreover, by homogeneity of degree zero, $x(p, w)$ depends only on the budget set the consumer faces. Hence $(\mathcal{B}^w, x(\cdot))$ is a choice structure, as defined in Section 1.C. Note that the choice structure $(\mathcal{B}^w, x(\cdot))$ does not include all possible subsets of X (e.g., it does not include all two- and three-element subsets of X). This fact will be significant for the relationship between the choice-based and preference-based approaches to consumer demand.

Comparative Statics

We are often interested in analyzing how the consumer's choice varies with changes in his wealth and in prices. The examination of a change in outcome in response to a change in underlying economic parameters is known as *comparative statics* analysis.

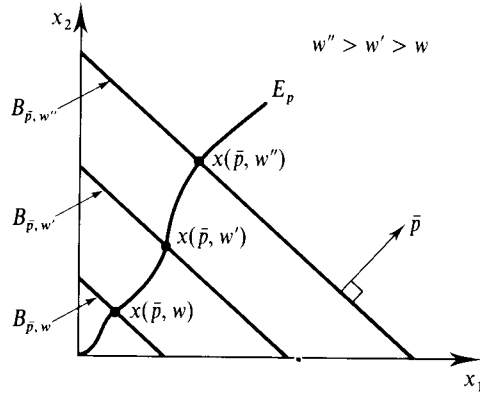
Wealth effects

For fixed prices $\bar{p}$, the function of wealth $x(\bar{p}, w)$ is called the consumer's *Engel function*. Its image in $\mathbb{R}_+^L$, $E_{\bar{p}} = \{x(\bar{p}, w): w > 0\}$, is known as the *wealth expansion path*. Figure 2.E.1 depicts such an expansion path.

At any (p, w) , the derivative $\partial x_\ell(p, w)/\partial w$ is known as the *wealth effect* for the ℓ th good.⁷

6. We use normalizations extensively in Part IV.

7. It is also known as the *income effect* in the literature. Similarly, the wealth expansion path is sometimes referred to as an *income expansion path*.

**Figure 2.E.1**

The wealth expansion path at prices $\bar{p}$.

A commodity ℓ is *normal* at (p, w) if $\partial x_\ell(p, w)/\partial w \geq 0$; that is, demand is nondecreasing in wealth. If commodity ℓ 's wealth effect is instead negative, then it is called *inferior* at (p, w) . If every commodity is normal at all (p, w) , then we say that *demand is normal*.

The assumption of normal demand makes sense if commodities are large aggregates (e.g., food, shelter). But if they are very disaggregated (e.g., particular kinds of shoes), then because of substitution to higher-quality goods as wealth increases, goods that become inferior at some level of wealth may be the rule rather than the exception.

In matrix notation, the wealth effects are represented as follows:

$$D_w x(p, w) = \begin{bmatrix} \frac{\partial x_1(p, w)}{\partial w} \\ \frac{\partial x_2(p, w)}{\partial w} \\ \vdots \\ \frac{\partial x_L(p, w)}{\partial w} \end{bmatrix} \in \mathbb{R}^L.$$

Price effects

We can also ask how consumption levels of the various commodities change as prices vary.

Consider first the case where $L = 2$, and suppose we keep wealth and price p_1 fixed. Figure 2.E.2 represents the demand function for good 2 as a function of its own price p_2 for various levels of the price of good 1, with wealth held constant at amount w . Note that, as is customary in economics, the price variable, which here is the independent variable, is measured on the vertical axis, and the quantity demanded, the dependent variable, is measured on the horizontal axis. Another useful representation of the consumers' demand at different prices is the locus of points demanded in $\mathbb{R}_+^2$ as we range over all possible values of p_2 . This is known as an *offer curve*. An example is presented in Figure 2.E.3.

More generally, the derivative $\partial x_\ell(p, w)/\partial p_k$ is known as the *price effect of p_k* , the price of good k , on the demand for good ℓ . Although it may be natural to think that a fall in a good's price will lead the consumer to purchase more of it (as in

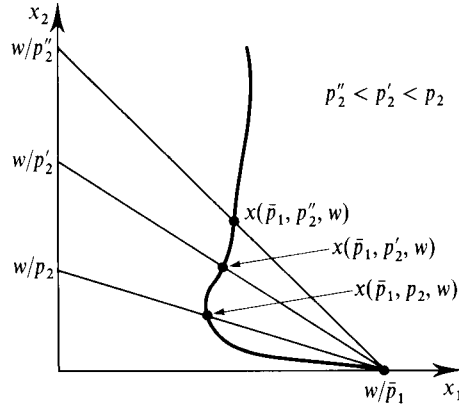
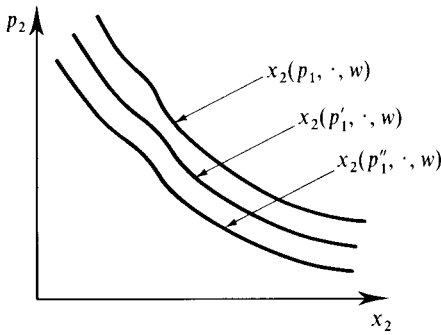


Figure 2.E.2 (top left)
The demand for good 2 as a function of its price (for various levels of p_1).

Figure 2.E.3 (top right)
An offer curve.

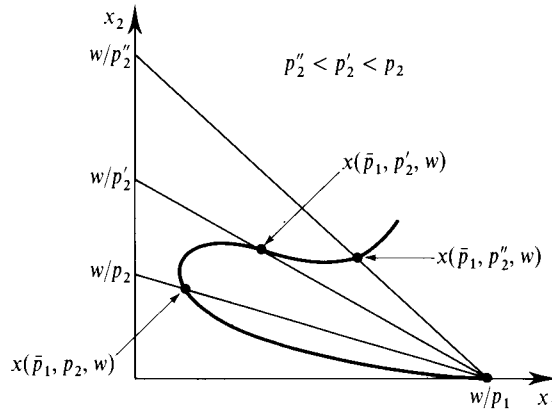


Figure 2.E.4 (bottom)
An offer curve where good 2 is inferior at $(\bar{p}, p'_2, w)$.

Figure 2.E.3), the reverse situation is not an economic impossibility. Good ℓ is said to be a *Giffen good* at (p, w) if $\partial x_\ell(p, w)/\partial p_\ell > 0$. For the offer curve depicted in Figure 2.E.4, good 2 is a Giffen good at $(\bar{p}_1, p'_2, w)$.

Low-quality goods may well be Giffen goods for consumers with low wealth levels. For example, imagine that a poor consumer initially is fulfilling much of his dietary requirements with potatoes because they are a low-cost way to avoid hunger. If the price of potatoes falls, he can then afford to buy other, more desirable foods that also keep him from being hungry. His consumption of potatoes may well fall as a result. Note that the mechanism that leads to potatoes being a Giffen good in this story involves a wealth consideration: When the price of potatoes falls, the consumer is effectively wealthier (he can afford to purchase more generally), and so he buys fewer potatoes. We will be investigating this interplay between price and wealth effects more extensively in the rest of this chapter and in Chapter 3.

The price effects are conveniently represented in matrix form as follows:

$$D_p x(p, w) = \begin{bmatrix} \frac{\partial x_1(p, w)}{\partial p_1} & \dots & \frac{\partial x_1(p, w)}{\partial p_L} \\ \vdots & \ddots & \vdots \\ \frac{\partial x_L(p, w)}{\partial p_1} & \dots & \frac{\partial x_L(p, w)}{\partial p_L} \end{bmatrix}.$$

Implications of homogeneity and Walras' law for price and wealth effects

Homogeneity and Walras' law imply certain restrictions on the comparative statics effects of consumer demand with respect to prices and wealth.

Consider, first, the implications of homogeneity of degree zero. We know that $x(\alpha p, \alpha w) - x(p, w) = 0$ for all $\alpha > 0$. Differentiating this expression with respect to α , and evaluating the derivative at $\alpha = 1$, we get the results shown in Proposition 2.E.1 (the result is also a special case of Euler's formula; see Section M.B of the Mathematical Appendix for details).

Proposition 2.E.1: If the Walrasian demand function $x(p, w)$ is homogeneous of degree zero, then for all p and w :

$$\sum_{k=1}^L \frac{\partial x_\ell(p, w)}{\partial p_k} p_k + \frac{\partial x_\ell(p, w)}{\partial w} w = 0 \text{ for } \ell = 1, \dots, L. \quad (2.E.1)$$

In matrix notation, this is expressed as

$$D_p x(p, w) p + D_w x(p, w) w = 0. \quad (2.E.2)$$

Thus, homogeneity of degree zero implies that the price and wealth derivatives of demand for any good ℓ , when weighted by these prices and wealth, sum to zero. Intuitively, this weighting arises because when we increase all prices and wealth proportionately, each of these variables changes in proportion to its initial level.

We can also restate equation (2.E.1) in terms of the *elasticities* of demand with respect to prices and wealth. These are defined, respectively, by

$$\varepsilon_{\ell k}(p, w) = \frac{\partial x_\ell(p, w)}{\partial p_k} \frac{p_k}{x_\ell(p, w)}$$

and

$$\varepsilon_{\ell w}(p, w) = \frac{\partial x_\ell(p, w)}{\partial w} \frac{w}{x_\ell(p, w)}.$$

These elasticities give the *percentage* change in demand for good ℓ per (marginal) percentage change in the price of good k or wealth; note that the expression for $\varepsilon_{\ell w}(\cdot, \cdot)$ can be read as $(\Delta x/x)/(\Delta w/w)$. Elasticities arise very frequently in applied work. Unlike the derivatives of demand, elasticities are independent of the units chosen for measuring commodities and therefore provide a unit-free way of capturing demand responsiveness.

Using elasticities, condition (2.E.1) takes the following form:

$$\sum_{k=1}^L \varepsilon_{\ell k}(p, w) + \varepsilon_{\ell w}(p, w) = 0 \text{ for } \ell = 1, \dots, L. \quad (2.E.3)$$

This formulation very directly expresses the comparative statics implication of homogeneity of degree zero: An equal percentage change in all prices and wealth leads to no change in demand.

Walras' law, on the other hand, has two implications for the price and wealth effects of demand. By Walras' law, we know that $p \cdot x(p, w) = w$ for all p and w . Differentiating this expression with respect to prices yields the first result, presented in Proposition 2.E.2.

Proposition 2.E.2: If the Walrasian demand function $x(p, w)$ satisfies Walras' law, then for all p and w :

$$\sum_{\ell=1}^L p_{\ell} \frac{\partial x_{\ell}(p, w)}{\partial p_k} + x_k(p, w) = 0 \quad \text{for } k = 1, \dots, L, \quad (2.E.4)$$

or, written in matrix notation,⁸

$$p \cdot D_p x(p, w) + x(p, w)^T = 0^T. \quad (2.E.5)$$

Similarly, differentiating $p \cdot x(p, w) = w$ with respect to w , we get the second result, shown in Proposition 2.E.3.

Proposition 2.E.3: If the Walrasian demand function $x(p, w)$ satisfies Walras' law, then for all p and w :

$$\sum_{\ell=1}^L p_{\ell} \frac{\partial x_{\ell}(p, w)}{\partial w} = 1, \quad (2.E.6)$$

or, written in matrix notation,

$$p \cdot D_w x(p, w) = 1. \quad (2.E.7)$$

The conditions derived in Propositions 2.E.2 and 2.E.3 are sometimes called the properties of *Cournot* and *Engel aggregation*, respectively. They are simply the differential versions of two facts: That total expenditure cannot change in response to a change in prices and that total expenditure must change by an amount equal to any wealth change.

Exercise 2.E.2: Show that equations (2.E.4) and (2.E.6) lead to the following two elasticity formulas:

$$\sum_{\ell=1}^L b_{\ell}(p, w) \varepsilon_{\ell k}(p, w) + b_k(p, w) = 0,$$

and

$$\sum_{\ell=1}^L b_{\ell}(p, w) \varepsilon_{\ell w}(p, w) = 1,$$

where $b_{\ell}(p, w) = p_{\ell} x_{\ell}(p, w)/w$ is the budget share of the consumer's expenditure on good ℓ given prices p and wealth w .

2.F The Weak Axiom of Revealed Preference and the Law of Demand

In this section, we study the implications of the weak axiom of revealed preference for consumer demand. Throughout the analysis, we continue to assume that $x(p, w)$ is single-valued, homogeneous of degree zero, and satisfies Walras' law.⁹

The weak axiom was already introduced in Section 1.C as a consistency axiom for the choice-based approach to decision theory. In this section, we explore its implications for the demand behavior of a consumer. In the preference-based approach to consumer behavior to be studied in Chapter 3, demand necessarily

8. Recall that 0^T means a row vector of zeros.

9. For generalizations to the case of multivalued choice, see Exercise 2.F.13.

satisfies the weak axiom. Thus, the results presented in Chapter 3, when compared with those in this section, will tell us how much more structure is imposed on consumer demand by the preference-based approach beyond what is implied by the weak axiom alone.¹⁰

In the context of Walrasian demand functions, the weak axiom takes the form stated in the Definition 2.F.1.

Definition 2.F.1: The Walrasian demand function $x(p, w)$ satisfies the *weak axiom of revealed preference* (the WA) if the following property holds for any two price–wealth situations (p, w) and (p', w') :

$$\text{If } p \cdot x(p', w') \leq w \text{ and } x(p', w') \neq x(p, w), \text{ then } p' \cdot x(p, w) > w'.$$

If you have already studied Chapter 1, you will recognize that this definition is precisely the specialization of the general statement of the weak axiom presented in Section 1.C to the context in which budget sets are Walrasian and $x(p, w)$ specifies a unique choice (see Exercise 2.F.1).

In the consumer demand setting, the idea behind the weak axiom can be put as follows: If $p \cdot x(p', w') \leq w$ and $x(p', w') \neq x(p, w)$, then we know that when facing prices p and wealth w , the consumer chose consumption bundle $x(p, w)$ even though bundle $x(p', w')$ was also affordable. We can interpret this choice as “revealing” a preference for $x(p, w)$ over $x(p', w')$. Now, we might reasonably expect the consumer to display some consistency in his demand behavior. In particular, given his revealed preference, we expect that he would choose $x(p, w)$ over $x(p', w')$ whenever they are both affordable. If so, bundle $x(p, w)$ must not be affordable at the price–wealth combination (p', w') at which the consumer chooses bundle $x(p', w')$. That is, as required by the weak axiom, we must have $p' \cdot x(p, w) > w'$.

The restriction on demand behavior imposed by the weak axiom when $L = 2$ is illustrated in Figure 2.F.1. Each diagram shows two budget sets $B_{p', w'}$ and $B_{p'', w''}$ and their corresponding choice $x(p', w')$ and $x(p'', w'')$. The weak axiom tells us that we cannot have both $p' \cdot x(p'', w'') \leq w'$ and $p'' \cdot x(p', w') \leq w''$. Panels (a) to (c) depict permissible situations, whereas demand in panels (d) and (e) violates the weak axiom.

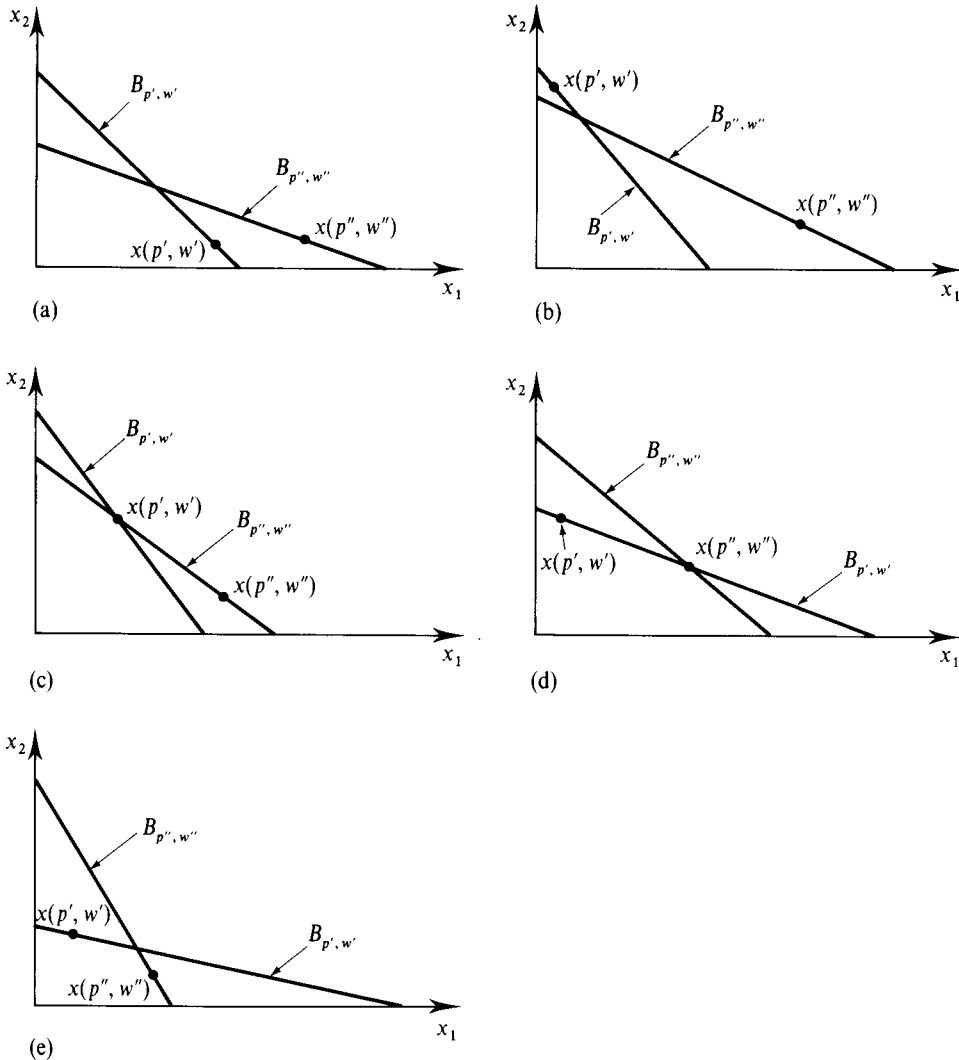
Implications of the Weak Axiom

The weak axiom has significant implications for the effects of price changes on demand. We need to concentrate, however, on a special kind of price change.

As the discussion of Giffen goods in Section 2.E suggested, price changes affect the consumer in two ways. First, they alter the relative cost of different commodities. But, second, they also change the consumer's real wealth: An increase in the price of a commodity impoverishes the consumers of that commodity. To study the implications of the weak axiom, we need to isolate the first effect.

One way to accomplish this is to imagine a situation in which a change in prices is accompanied by a change in the consumer's wealth that makes his initial consumption bundle just affordable at the new prices. That is, if the consumer is originally facing prices p and wealth w and chooses consumption bundle $x(p, w)$, then

10. Or, stated more properly, beyond what is implied by the weak axiom in conjunction with homogeneity of degree zero and Walras' law.

**Figure 2.F.1**

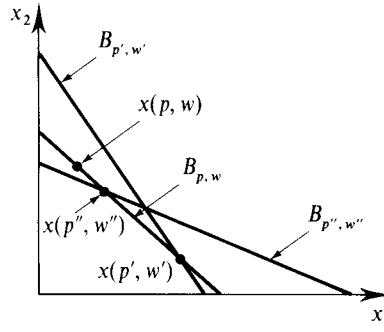
Demand in panels (a) to (c) satisfies the weak axiom; demand in panels (d) and (e) does not.

when prices change to p' , we imagine that the consumer's wealth is adjusted to $w' = p' \cdot x(p, w)$. Thus, the wealth adjustment is $\Delta w = \Delta p \cdot x(p, w)$, where $\Delta p = (p' - p)$. This kind of wealth adjustment is known as *Slutsky wealth compensation*. Figure 2.F.2 shows the change in the budget set when a reduction in the price of good 1 from p_1 to p'_1 is accompanied by Slutsky wealth compensation. Geometrically, the restriction is that the budget hyperplane corresponding to (p', w') goes through the vector $x(p, w)$.

We refer to price changes that are accompanied by such compensating wealth changes as *(Slutsky) compensated price changes*.

In Proposition 2.F.1, we show that the weak axiom can be equivalently stated in terms of the demand response to compensated price changes.

Proposition 2.F.1: Suppose that the Walrasian demand function $x(p, w)$ is homogeneous of degree zero and satisfies Walras' law. Then $x(p, w)$ satisfies the weak axiom if and only if the following property holds:

**Figure 2.F.3**

The weak axiom holds if and only if it holds for all compensated price changes.

Now choose the value of $\alpha \in (0,1)$ for which

$$(\alpha p' + (1 - \alpha)p'') \cdot x(p', w') = (\alpha p' + (1 - \alpha)p'') \cdot x(p'', w''),$$

and denote $p = \alpha p' + (1 - \alpha)p''$ and $w = (\alpha p' + (1 - \alpha)p'') \cdot x(p', w')$. This construction is illustrated in Figure 2.F.3. We then have

$$\begin{aligned} \alpha w' + (1 - \alpha)w'' &> \alpha p' \cdot x(p', w') + (1 - \alpha)p'' \cdot x(p', w') \\ &= w \\ &= p \cdot x(p, w) \\ &= \alpha p' \cdot x(p, w) + (1 - \alpha)p'' \cdot x(p, w). \end{aligned}$$

Therefore, either $p' \cdot x(p, w) < w'$ or $p'' \cdot x(p, w) < w''$. Suppose that the first possibility holds (the argument is identical if it is the second that holds). Then we have $x(p, w) \neq x(p', w')$, $p \cdot x(p', w') = w$, and $p' \cdot x(p, w) < w'$, which constitutes a violation of the weak axiom for the compensated price change from (p', w') to (p, w) .

Once we know that in order to test for the weak axiom it suffices to consider only compensated price changes, the remaining reasoning is straightforward. If the weak axiom does not hold, there exists a compensated price change from some (p', w') to some (p, w) such that $x(p, w) \neq x(p', w')$, $p \cdot x(p', w') = w$, and $p' \cdot x(p, w) \leq w'$. But since $x(\cdot, \cdot)$ satisfies Walras' law, these two inequalities imply

$$p \cdot [x(p', w') - x(p, w)] = 0 \quad \text{and} \quad p' \cdot [x(p', w') - x(p, w)] \geq 0.$$

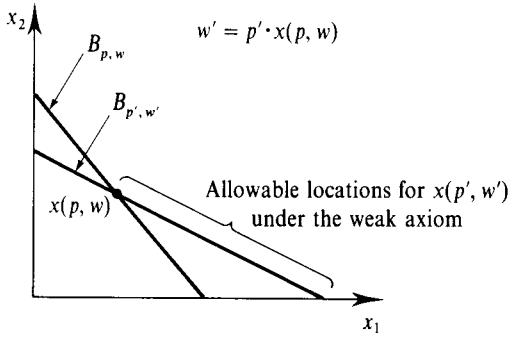
Hence, we would have

$$(p' - p) \cdot [x(p', w') - x(p, w)] \geq 0 \quad \text{and} \quad x(p, w) \neq x(p', w'),$$

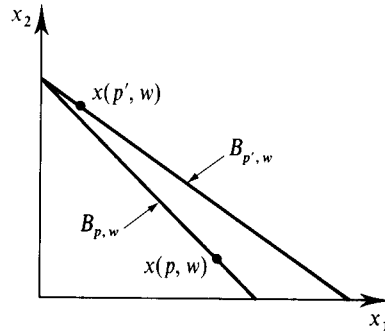
which is a contradiction to (2.F.1) holding for all compensated price changes [and with strict inequality when $x(p, w) \neq x(p', w')$]. ■

The inequality (2.F.1) can be written in shorthand as $\Delta p \cdot \Delta x \leq 0$, where $\Delta p = (p' - p)$ and $\Delta x = [x(p', w') - x(p, w)]$. It can be interpreted as a form of the *law of demand*: *Demand and price move in opposite directions*. Proposition 2.F.1 tells us that the law of demand holds for *compensated* price changes. We therefore call it the *compensated law of demand*.

The simplest case involves the effect on demand for some good ℓ of a compensated change in its own price p_ℓ . When only this price changes, we have $\Delta p = (0, \dots, 0, \Delta p_\ell, 0, \dots, 0)$. Since $\Delta p \cdot \Delta x = \Delta p_\ell \Delta x_\ell$, Proposition 2.F.1 tells us that if $\Delta p_\ell > 0$, then we must have $\Delta x_\ell < 0$. The basic argument is illustrated in Figure 2.F.4. Starting at

**Figure 2.F.4 (left)**

Demand must be nonincreasing in own price for a compensated price change.

**Figure 2.F.5 (right)**

Demand for good 1 can fall when its price decreases for an uncompensated price change.

(p, w) , a compensated decrease in the price of good 1 rotates the budget line through $x(p, w)$. The WA allows moves of demand only in the direction that increases the demand of good 1.

Figure 2.F.5 should persuade you that the WA (or, for that matter, the preference maximization assumption discussed in Chapter 3) is not sufficient to yield the law of demand for price changes that are *not* compensated. In the figure, the price change from p to p' is obtained by a decrease in the price of good 1, but the weak axiom imposes no restriction on where we place the new consumption bundle; as drawn, the demand for good 1 falls.

When consumer demand $x(p, w)$ is a differentiable function of prices and wealth, Proposition 2.F.1 has a differential implication that is of great importance. Consider, starting at a given price–wealth pair (p, w) , a differential change in prices dp . Imagine that we make this a compensated price change by giving the consumer compensation of $dw = x(p, w) \cdot dp$ [this is just the differential analog of $\Delta w = x(p, w) \cdot \Delta p$]. Proposition 2.F.1 tells us that

$$dp \cdot dx \leq 0. \quad (2.F.5)$$

Now, using the chain rule, the differential change in demand induced by this compensated price change can be written as

$$dx = D_p x(p, w) dp + D_w x(p, w) dw. \quad (2.F.6)$$

Hence

$$dx = D_p x(p, w) dp + D_w x(p, w) [x(p, w) \cdot dp] \quad (2.F.7)$$

or equivalently

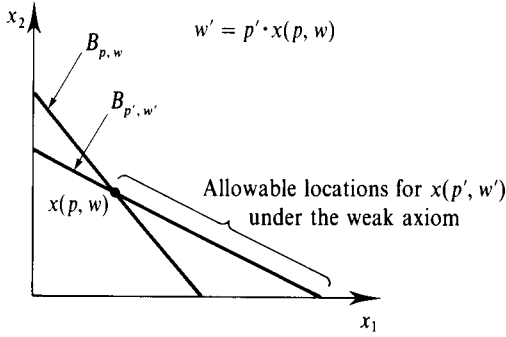
$$dx = [D_p x(p, w) + D_w x(p, w) x(p, w)^T] dp. \quad (2.F.8)$$

Finally, substituting (2.F.8) into (2.F.5) we conclude that for any possible differential price change dp , we have

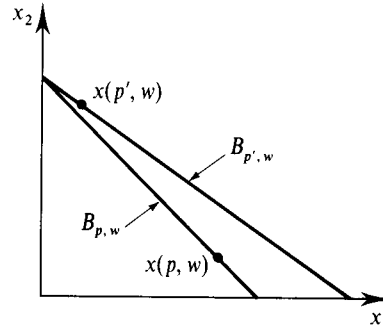
$$dp \cdot [D_p x(p, w) + D_w x(p, w) x(p, w)^T] dp \leq 0. \quad (2.F.9)$$

The expression in square brackets in condition (2.F.9) is an $L \times L$ matrix, which we denote by $S(p, w)$. Formally

$$S(p, w) = \begin{bmatrix} s_{11}(p, w) & \cdots & s_{1L}(p, w) \\ \vdots & \ddots & \vdots \\ s_{L1}(p, w) & \cdots & s_{LL}(p, w) \end{bmatrix},$$

**Figure 2.F.4 (left)**

Demand must be nonincreasing in own price for a compensated price change.

**Figure 2.F.5 (right)**

Demand for good 1 can fall when its price decreases for an uncompensated price change.

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When consumer demand $x(p, w)$ is a differentiable function of prices and wealth, Proposition 2.F.1 has a differential implication that is of great importance. Consider, starting at a given price–wealth pair (p, w) , a differential change in prices dp . Imagine that we make this a compensated price change by giving the consumer compensation of $dw = x(p, w) \cdot dp$ [this is just the differential analog of $\Delta w = x(p, w) \cdot \Delta p$]. Proposition 2.F.1 tells us that

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Now, using the chain rule, the differential change in demand induced by this compensated price change can be written as

$$dx = D_p x(p, w) dp + D_w x(p, w) dw. \quad (2.F.6)$$

Hence

$$dx = D_p x(p, w) dp + D_w x(p, w) [x(p, w) \cdot dp] \quad (2.F.7)$$

or equivalently

$$dx = [D_p x(p, w) + D_w x(p, w) x(p, w)^T] dp. \quad (2.F.8)$$

Finally, substituting (2.F.8) into (2.F.5) we conclude that for any possible differential price change dp , we have

$$dp \cdot [D_p x(p, w) + D_w x(p, w) x(p, w)^T] dp \leq 0. \quad (2.F.9)$$

The expression in square brackets in condition (2.F.9) is an $L \times L$ matrix, which we denote by $S(p, w)$. Formally

$$S(p, w) = \begin{bmatrix} s_{11}(p, w) & \cdots & s_{1L}(p, w) \\ \vdots & \ddots & \vdots \\ s_{L1}(p, w) & \cdots & s_{LL}(p, w) \end{bmatrix},$$

where the (ℓ, k) th entry is

$$s_{\ell k}(p, w) = \frac{\partial x_{\ell}(p, w)}{\partial p_k} + \frac{\partial x_{\ell}(p, w)}{\partial w} x_k(p, w). \quad (2.F.10)$$

The matrix $S(p, w)$ is known as the *substitution*, or *Slutsky*, matrix, and its elements are known as *substitution effects*.

The “substitution” terminology is apt because the term $s_{\ell k}(p, w)$ measures the differential change in the consumption of commodity ℓ (i.e., the substitution to or from other commodities) due to a differential change in the price of commodity k when wealth is adjusted so that the consumer can still just afford his original consumption bundle (i.e., due solely to a change in relative prices). To see this, note that the change in demand for good ℓ if wealth is left unchanged is $(\partial x_{\ell}(p, w)/\partial p_k) dp_k$. For the consumer to be able to “just afford” his original consumption bundle, his wealth must vary by the amount $x_k(p, w) dp_k$. The effect of this wealth change on the demand for good ℓ is then $(\partial x_{\ell}(p, w)/\partial w) [x_k(p, w) dp_k]$. The sum of these two effects is therefore exactly $s_{\ell k}(p, w) dp_k$.

We summarize the derivation in equations (2.F.5) to (2.F.10) in Proposition 2.F.2.

Proposition 2.F.2: If a differentiable Walrasian demand function $x(p, w)$ satisfies Walras’ law, homogeneity of degree zero, and the weak axiom, then at any (p, w) , the Slutsky matrix $S(p, w)$ satisfies $v \cdot S(p, w) v \leq 0$ for any $v \in \mathbb{R}^L$.

A matrix satisfying the property in Proposition 2.F.2 is called *negative semidefinite* (it is *negative definite* if the inequality is strict for all $v \neq 0$). See Section M.D of the Mathematical Appendix for more on these matrices.

Note that $S(p, w)$ being negative semidefinite implies that $s_{\ell \ell}(p, w) \leq 0$: That is, *the substitution effect of good ℓ with respect to its own price is always nonpositive*.

An interesting implication of $s_{\ell \ell}(p, w) \leq 0$ is that a good can be a Giffen good at (p, w) only if it is inferior. In particular, since

$$s_{\ell \ell}(p, w) = \partial x_{\ell}(p, w)/\partial p_{\ell} + [\partial x_{\ell}(p, w)/\partial w] x_{\ell}(p, w) \leq 0,$$

if $\partial x_{\ell}(p, w)/\partial p_{\ell} > 0$, we must have $\partial x_{\ell}(p, w)/\partial w < 0$.

For later reference, we note that Proposition 2.F.2 does *not* imply, in general, that the matrix $S(p, w)$ is symmetric.¹¹ For $L = 2$, $S(p, w)$ is necessarily symmetric (you are asked to show this in Exercise 2.F.11). When $L > 2$, however, $S(p, w)$ need not be symmetric under the assumptions made so far (homogeneity of degree zero, Walras’ law, and the weak axiom). See Exercises 2.F.10 and 2.F.15 for examples. In Chapter 3 (Section 3.H), we shall see that the symmetry of $S(p, w)$ is intimately connected with the possibility of generating demand from the maximization of rational preferences.

Exploiting further the properties of homogeneity of degree zero and Walras’ law, we can say a bit more about the substitution matrix $S(p, w)$.

11. A matter of terminology: It is common in the mathematical literature that “definite” matrices are assumed to be symmetric. Rigorously speaking, if no symmetry is implied, the matrix would be called “quasidefinite.” To simplify terminology, we use “definite” without any supposition about symmetry; if a matrix is symmetric, we say so explicitly. (See Exercise 2.F.9.)

Proposition 2.F.3: Suppose that the Walrasian demand function $x(p, w)$ is differentiable, homogeneous of degree zero, and satisfies Walras' law. Then $p \cdot S(p, w) = 0$ and $S(p, w)p = 0$ for any (p, w) .

Exercise 2.F.7: Prove Proposition 2.F.3. [*Hint:* Use Propositions 2.E.1 to 2.E.3.]

It follows from Proposition 2.F.3 that the matrix $S(p, w)$ is always singular (i.e., it has rank less than L), and so the negative semidefiniteness of $S(p, w)$ established in Proposition 2.F.2 cannot be extended to negative definiteness (e.g., see Exercise 2.F.17).

Proposition 2.F.2 establishes negative semidefiniteness of $S(p, w)$ as a necessary implication of the weak axiom. One might wonder: Is this property sufficient to imply the WA [so that negative semidefiniteness of $S(p, w)$ is actually equivalent to the WA]? That is, if we have a demand function $x(p, w)$ that satisfies Walras' law, homogeneity of degree zero and has a negative semidefinite substitution matrix, must it satisfy the weak axiom? The answer is *almost, but not quite*. Exercise 2.F.16 provides an example of a demand function with a negative semidefinite substitution matrix that violates the WA. The sufficient condition is that $v \cdot S(p, w)v < 0$ whenever $v \neq \alpha p$ for any scalar α ; that is, $S(p, w)$ must be negative definite for all vectors other than those that are proportional to p . This result is due to Samuelson [see Samuelson (1947) or Kihlstrom, Mas-Colell, and Sonnenschein (1976) for an advanced treatment]. The gap between the necessary and sufficient conditions is of the same nature as the gap between the necessary and the sufficient second-order conditions for the minimization of a function.

Finally, how would a theory of consumer demand that is based solely on the assumptions of homogeneity of degree zero, Walras' law, and the consistency requirement embodied in the weak axiom compare with one based on rational preference maximization?

Based on Chapter 1, you might hope that Proposition 1.D.2 implies that the two are equivalent. But we cannot appeal to that proposition here because the family of Walrasian budgets does not include every possible budget; in particular, it does not include all the budgets formed by only two- or three-commodity bundles.

In fact, the two theories are not equivalent. For Walrasian demand functions, the theory derived from the weak axiom is weaker than the theory derived from rational preferences, in the sense of implying fewer restrictions. This is shown formally in Chapter 3, where we demonstrate that if demand is generated from preferences, or is capable of being so generated, then it must have a symmetric Slutsky matrix at all (p, w) . But for the moment, Example 2.F.1, due originally to Hicks (1956), may be persuasive enough.

Example 2.F.1: In a three-commodity world, consider the three budget sets determined by the price vectors $p^1 = (2, 1, 2)$, $p^2 = (2, 2, 1)$, $p^3 = (1, 2, 2)$ and wealth = 8 (the same for the three budgets). Suppose that the respective (unique) choices are $x^1 = (1, 2, 2)$, $x^2 = (2, 1, 2)$, $x^3 = (2, 2, 1)$. In Exercise 2.F.2, you are asked to verify that any two pairs of choices satisfy the WA but that x^3 is revealed preferred to x^2 , x^2 is revealed preferred to x^1 , and x^1 is revealed preferred to x^3 . This situation is incompatible with the existence of underlying rational preferences (transitivity would be violated).